

The split-zero programme around the Riemann zeta function: an audited record of its results, with proofs

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Abstract

The split-zero programme of KokunoYumeto studies the nontrivial zeros of the Riemann zeta function through the Hurwitz sheets $\zeta(s, 1+t)$, along which the zeros move; through quotient modules of entire functions attached to the zeros, above all Meyer's quotient $\mathcal{B}/I_\zeta$; and through arithmetic, cohomological and positivity constructions modelled on Weil's explicit formula and on Deligne's proof of the Weil conjectures. This record states what the programme and an audit of it have established, with proofs. The primary decomposition of Meyer's quotient converges for every class if and only if the principal parts of $1/\zeta$ at the zeros are polynomially bounded; the Nyman–Beurling family is dense in the Fréchet topology unconditionally, already for the degrees 2^{a3^b} when $t > (\log 2)(\log 3)/(4\pi)$; among the forms compatible with the scaling transfer, positivity is exactly the condition that removes the off-line zeros; in a two-line system the shifted zero set is the mirror of the zero set, Weil's quadratic form sits on the diagonal of a hyperbolic chiral form, the holonomy index is Turing's index, and the two-line quotient splits for every primitive Dirichlet L-function; a formal logarithm detects freeness of monoids of integers and characterizes the Eulerian sheets, and the even lattice sums with $\varphi(q) > 2$ have infinitely many zeros on both sides of the critical line; and the programme's exponent data have tensor-power abscissa exactly $1 + k \cdot \max \operatorname{Re} \rho$, an algebraic purity criterion sits at the core of Weil II 1.8.4, and thirteen misprints of the printed Weil II are recorded. Proved negative results are stated with their scope and the input each construction would need, and the bridges to other areas are described. No result here proves or disproves the Riemann hypothesis.

0. Introduction

0.1 Why this record

The Riemann hypothesis (RH) asserts that every nontrivial zero of ζ has real part $1/2$. Its classical reformulations include Weil's positivity criterion for the explicit formula [We52, Bo], the Nyman–Beurling criterion [Be, BD], and the analogy with curves over finite fields, where RH is Weil's theorem [We48] and whose higher-dimensional form is Deligne's [De]. The programme builds constructions around these reformulations and around the flow of the Hurwitz sheets $\zeta(s, 1+t)$, which moves the zeros of ζ off the critical line. It consists of several hundred numbered statements with proofs, with labels such as MCL5, OMS6 or GSL4, published in the Zenodo record *Split-Zero cohomology and the zeta-function research programme* (concept DOI 10.5281/zenodo.22678085) and in the repository KokunoYumeto/zeta-function-research-reader. This record states what holds, with proofs or with the location of the proof, what fails and in which scope, and how the programme connects with other areas, so that a reader need not work through the files to find out.

0.2 Main results

Results marked *new* were found in the audit; the others are the programme's, checked in the audit.

- **Meyer's quotient** (§3). **Theorem 3.9** (*new*): the primary decomposition of $Q = \mathcal{B}/I_\zeta$ converges for every class if and only if the principal parts of $1/\zeta$ at the zeros are polynomially bounded, if and only if the jet map is an isomorphism onto the rapidly decreasing jet tuples; with the spectral theory of multiplication by s (Proposition 3.2) and the exact summation image (Proposition 3.5).
- **Density and positivity** (§3). Nyman–Beurling density in $\mathcal{B}$, unconditional, and under two Carleman criteria (*new*) already for 2^{a3^b} when $t > 0.06060\dots$ (Proposition 3.11), with Proposition 3.12 (*new*); the transfer-compatible Hermitian forms, positive exactly when supported on the critical zeros (Lemma 3.13; the Hermitian classification *new*).

- **The two lines** (§4, *new*). The first Hurwitz jet and the mirror set, with mirror equal to shift at every zero if and only if RH holds (Theorems 4.1–4.2); Weil’s form on the diagonal of the chiral cross form (Proposition 4.6, Corollary 4.7); the holonomy index as Turing’s index (Proposition 4.9); the splitting for every primitive Dirichlet L-function (Proposition 4.12).
- **Sheets and monoids** (§2). The formal logarithm detects freeness (Theorem 2.4, stronger form *new*; Corollary 2.5, *new*); $Z_{1/q}$ is Eulerian exactly for $q \in \{3, 4, 6\}$ and has infinitely many zeros on both sides of the line when $\varphi(q) > 2$ (Propositions 2.8–2.9, *new*); the velocity results of §2.1 (*new*).
- **Weights** (§5). Propositions 5.1 (*new*), 5.2 and 5.3, Lemma 5.4 (*new*, a corrected constant in Weil II 2.1.7), and thirteen misprints in print (§5.4).
- **Part B** (§8): thirty proved negative results, each with its scope and the input the construction would need. **Part C** (§9): the bridges.

0.3 What the results mean

The quotient $Q = \mathcal{B}/I_\zeta$ is the programme’s central object, and §3 describes it completely as a module under multiplication by s . Theorem 3.9 identifies the one structural question left, whether Q is the convergent sum of its primary parts, with polynomial lower bounds for the principal parts of $1/\zeta$ at the zeros; that question is open, and a positive answer would imply that every zero is simple.

Every bounded positive form compatible with the scaling transfer is supported on the critical zeros, while an off-line pair enters such a form as a hyperbolic block (Lemma 3.13). The two-line system of §4 realizes this geometrically: Weil’s form sits on the diagonal of a chiral form that is hyperbolic for every configuration of zeros, and counting in the system is Turing’s method.

The Euler product is located precisely: the Eulerian sheets are $a \in \{1, 1/2\}$, the even lattice sums with $\varphi(q) > 2$ have zeros off the line, and the Davenport–Heilbronn function shares the functional equation and the two-line structure of ζ while having a zero off the line. A construction whose proof uses only properties shared with these functions has conclusions that hold for them too, and Part B records the input that would distinguish ζ , in most cases the Euler product. On Deligne’s side, the uniform pole bound complementing the programme’s positivity step is equivalent on its finite data to $\max \operatorname{Re} \rho \leq 1/2$ (Proposition 5.1), so it has to come from an independent source, as the H^2_c of a fixed curve does in Weil II (Question 8). No result here proves or disproves RH, and none is presented as doing so.

0.4 How this record was made

This record is an audit, carried out by Claude between 22 and 26 September 2026, of the programme’s published work: the second attempt’s proofs, the continuation bulletins of 24–25 September, the programme’s reconstruction of Weil II and its Deligne reader, the final reports of its working sessions of 24–25 September, and the material of the author’s other workbenches used here (§11 lists what was not read). The programme, its constructions and most of its theorems are the work of its author, KokunoYumeto, developed with ChatGPT and Codex (OpenAI). Programme statements are cited by their labels, which the programme’s reading guide 181-reading-guide-and-file-locations.md and its index 182-complete-file-and-migration-index.json locate; results found in the audit are credited as such.

Every block was read against its own proofs. The audit’s notes were each refereed by an instance that had not written them (nineteen passes and a two-part final pass), the reader compiled from them had its own referee pass, and a later review looked for understatement and missed generality as well as for overstatement; every finding was re-derived before it was applied. Every statement carries one of four statuses: **verified** (proved here; or proved in the programme at the cited label and checked in the audit; or re-computed by a named check); **not verified here** (nothing further is implied, in either direction); **my assessment** (a view with its reasons, in the first person); **proved negative** (a statement shown false or limited, with the counterexample or derivation printed). A statement supported by computation alone is labelled as numerical.

Re-deriving the withdrawn texts changed several statements, and those given here are the corrected ones (Proposition 5.2 at on-line zeros; Corollary 2.3 for $a^\zeta(s, a)$, since $\zeta(s, 1/2) = 2^s \prod_{p>2} (1 - p^{-s})^{-1}$; Theorem 2.2(a); step 4 of Proposition 3.11). An earlier claim made in the audit, that the

zeros of $\zeta(s, 1+t)$ lie on vertical lines only at the Eulerian times, is false: zero 4 of ζ crosses $\text{Re } s = 1/2$ at $t = 0.95095300425$ and zero 5 at $t = 0.858704259442$ (numerical, to 25 digits).

0.5 Organization

Part A (§§1–7) contains the verified results: the setting (§1), sheets, monoids and Euler products (§2), Meyer’s quotient (§3), the two lines (§4), weights (§5), further results (§6) and a catalogue of the remaining results (§7). Part B (§8) contains the proved negative results, Part C (§9) the bridges and the directions they open, §10 the questions and §11 the verification.

Part A. Verified results

1. The setting

The programme’s idea. The programme’s coefficient algebra keeps a ring’s internal zero apart from a separate symbol τ for absence (the split-zero semiring $G(R) = R \sqcup \{\tau\}$), so that a zero element in a support fibre is not confused with an absent fibre. On the zeta side it follows the zeros of ζ along the Hurwitz sheets $t \mapsto \zeta(s, 1+t)$ and asks which constructions (quotient modules, receivers, positivity forms, cohomology over a reconstructed arithmetic) determine where the zeros lie. In the author’s picture the support τ is the pivot of the swing between 0 and 1; on the zeta side the corresponding pivot is the reflection $s \mapsto 1 - \bar{s}$ about $\text{Re } s = 1/2$ (a proposed typed matching, §9.8). We write $\bar{w}$ or $\text{conj}(w)$ for the complex conjugate.

Sheets and zeros. For $a > 0$, $\zeta(s, a) = \sum_{n \geq 0} (n+a)^{-s}$ on $\text{Re } s > 1$, continued meromorphically with one simple pole at $s = 1$; the flow is $t \mapsto \zeta(s, 1+t)$, the Eulerian sheets are $\zeta(s, 1) = \zeta(s)$ and $\zeta(s, 1/2) = (2^s - 1)\zeta(s)$, and the even lattice sums are $Z_a(s) = \zeta(s, a) + \zeta(s, 1-a)$, $0 < a < 1$. Put $\xi(s) = (1/2)s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ and $F_*(s) = \xi(s)/4$ (the programme’s F_0). $\mathcal{Z}$ is the set of distinct nontrivial zeros $\rho = \beta + i\gamma$, with multiplicities m_ρ written out in sums, and $\rho^\# = 1 - \bar{\rho}$; since $\xi(1-s) = \xi(s)$ and $\xi(\bar{s}) = \text{conj}(\xi(s))$, $\#$ permutes $\mathcal{Z}$ preserving multiplicities, and RH says exactly that $\rho^\# = \rho$ for every ρ . For $T > 0$ not an ordinate, $N(T)$ counts the zeros with $0 < \gamma \leq T$, $N_0(T)$ those on the line and $P(T)$ those with $\beta < 1/2$ (with multiplicity); Hardy’s function $Z(t) = e^{i\theta(t)}\zeta(1/2 + it)$ is real, and $V(T)$ is its number of sign changes on $(0, T)$.

The strip space and Meyer’s quotient. $\mathcal{B}$ is the Fréchet space of entire F with $b_{\{A,M\}}(F) = \sup_{t \in \mathbb{R}} \{|\sigma| \leq A, t \in \mathbb{R}\} (1 + |t|)^M |F(\sigma + it)| < \infty$ for all $A, M \geq 0$. For a divisor D , $I_D \subset \mathcal{B}$ is the closed ideal of functions vanishing to order at least $D(\lambda)$ at each λ , and $I_\mathcal{Z} = I_{\mathcal{Z}}$. The programme’s Meyer quotient is $Q = \mathcal{A}/\mathcal{E}S \cong \mathcal{B}/I_\mathcal{Z}$ (OMS3.14; [Me]), where $\mathcal{E}$ is Connes’ summation map and the isomorphism is the Mellin transform of the exact summation image (Proposition 3.5); L denotes multiplication by s . With T a formal variable, the jet at ρ is $j_\rho[F] = F(\rho + T) \bmod T^{\{m_\rho\}} \in A_\rho = \mathbb{C}[T]/(T^{\{m_\rho\}})$, and the primary projector is $\Pi_\rho = s_{\rho j_\rho}$ with the programme’s sections s_ρ (OMS6).

Two lines, three maps and Weil’s pairing. The two-line function $G(s) = F_*(s)F_*(s+1) = \xi(s)\xi(s+1)/16$ has zero set $\mathcal{Z} \sqcup (\mathcal{Z} - 1)$, in $0 < \text{Re } s < 1$ and $-1 < \text{Re } s < 0$. The maps are the reflection $s^\# = 1 - \bar{s}$ through $1/2$, the mirror $A(s) = -\bar{s}$ through the pole at 0, and the translation T between the lines; on $\mathcal{Z}$, $A(\rho) = \rho^\# - 1$, that is $T^{\{-1\}}A = \#$. For finitely supported u on $\mathcal{Z}$, Weil’s pairing is $\Omega(u) = \sum_\rho m_\rho u(\rho) \text{conj}(u(\rho^\#))$, the zero side of Weil’s quadratic form [We52, Bo]; Weil’s criterion says that RH holds if and only if Weil’s functional is nonnegative on every $g * g^\#$, and $\Omega \geq 0$ on all finitely supported u exactly when every zero is on the line (Proposition 4.6(b)).

2. Hurwitz sheets, monoids of integers and Euler products

2.1 Zero velocities on the sheets

Along the flow $a = 1 + t$ a simple zero ρ of $\zeta(s, a)$ moves with velocity $c(\rho) = \rho \zeta(\rho + 1, a) / \partial_s \zeta(\rho, a)$, by implicit differentiation from $\partial_a \zeta(s, a) = -s \zeta(s + 1, a)$ (Theorem 4.1). At $a = 1$ the first zero has velocity $0.98358 + 9.65780i$ and leaves the line to the right as t increases; numerically the first five zeros leave the

line for $t > 0$, and zeros 4 and 5 cross back at the non-Eulerian times $t \approx 0.95095$ and $t \approx 0.85870$. The following results were found in the audit (*new*).

Theorem 2.1 (the velocity sums). Let $a > 0$, let c_0 exceed the real part of every zero of $\zeta(s, a)$, and let h be entire with $(1 + |\operatorname{Re} z|)^A h(z)$ bounded on $|\operatorname{Im} z| \leq c_0 + 2$ for every A ; put $H(s) = h(-i(s - 1/2))$ and $\hat{h}_1(\xi) = (2\pi)^{-1} \int_{1/2 + iu}^{1/2 + i\xi} h(u) e^{-iu\xi} du$. Put $F_a(s) = s\zeta(s + 1, a)/\zeta(s, a)$ and $a\zeta(s + 1, a)/\zeta(s, a) = \sum_r \beta_r(a) r^{-s}$, r in the multiplicative monoid generated by the numbers $1 + n/a$ ($\beta_1 = 1$). Along good heights $T_k \rightarrow \infty$ (where $1/\zeta(\sigma \pm iT_k, a)$ is polynomially bounded on the horizontal segments),

$$\lim_k \sum_{|\gamma| < T_k} \operatorname{Res}_\rho(F_a H) = (1/a)(2\pi)^{-1} \int_{1/2 + iu}^{1/2 + i\infty} h(u) du + (1/a) \sum_{r > 1} \beta_r(a) r^{-1/2} \hat{h}_1(\log r) + L_a(h),$$

with the explicit left-line term $L_a(h) = -(2\pi i)^{-1} \int_{\operatorname{Re} s = -b} F_a H ds$ ($3/2 < b < 2$).

Theorem 2.2 (consequences). (a) The smoothed sums of $\operatorname{Re} c$ and $\operatorname{Im} c$ over $T_1 < \gamma \leq T_2$, $10\Delta \leq T_1$, grow like $(T_2 - T_1)/(4\pi a)$ and $(T_2^2 - T_1^2)/(4\pi a) - (T_2 - T_1)$; for fixed width Δ the horizontal error is of order T , and at $a = 1$ both errors tend to 0 when $\Delta \geq (1 + \delta)\sqrt{(2 \log T_2)/\log 2}$ and $T_1 \geq \max(10\Delta, 4\Delta\sqrt{(\log \Delta)})$, given constants independent of Δ . (b) Under (H), all but finitely many zeros simple and on the line, the sharp laws have errors that are unbounded multiples of T (normalized mean square with $\limsup \geq 0.0373$); the almost periodicity of the smoothed error is unconditional. (c) The smoothed spectrum has frequencies $\log r$ and amplitudes $-\beta_r(a) r^{-1/2} e^{-\Delta^2 \log^2 r / 2} / (2\pi a \log r)$. (d) The velocity spectrum factors over a $\mathbb{Q}$ -independent basis if and only if $a \in \{1, 1/2\}$, where the primes are exactly the frequencies of maximal amplitude. (e) If the zeros up to γ_n are simple and on the line (true for $n \leq 1.2 \cdot 10^{13}$ [PT]), $\arg c(\rho_n) = \pi/2 - \arctan(1/(2\gamma_n)) + \pi(S_{\{3/2\}}(\gamma_n) - S_n)$, $S_n = n - 3/2 - \theta(\gamma_n)/\pi$, $S_{\{3/2\}}(t) = \pi^{-1} \arg \zeta(3/2 + it)$.

Corollary 2.3 (only two sheets are Eulerian). The normalized sheet $a^\wedge s \zeta(s, a)$ has an Euler product with local factors $(1 - P^\wedge \{-s\})^{-1}$ only for $a \in \{1, 1/2\}$.

Status of 2.1–2.3. Verified in the audit: the proofs were checked as stated (2.2(d) with a full referee pass; 2.2(b) completed in the audit; 2.3 follows from 2.2(d)), and Theorem 2.1 was also checked numerically at $a = 1$ (relative difference $1.8 \cdot 10^{-18}$). The proofs are not reproduced in this record. At $a = 1/q$, 2.2(d) is the factoriality criterion for congruence monoids [BC, Theorem 3.4(2)] (§2.3).

2.2 Monoids of integers and the formal logarithm

Let M be a multiplicative submonoid of the positive integers. An atom is an $m > 1$ in M that is not a product of two elements of $M \setminus \{1\}$; M is free if every element has exactly one factorization into atoms, and half-factorial if all factorizations of each element have the same length. With $D_M(s) = \sum_{m \in M} m^{-s}$, $= 1 + E(s)$, put $\log D_M = \sum_{k \geq 1} ((-1)^{k+1}/k) E^k = \sum_{n \geq 2} r_n n^{-s}$, so that $r_n = \sum_k ((-1)^{k+1}/k) T_k(n)$, where $T_k(n)$ counts the ordered k -tuples in $(M \setminus \{1\})^k$ with product n .

Theorem 2.4 (the formal logarithm detects freeness). Let $n \in M$ and suppose that every proper M -divisor of n (an $m \in M$ with $1 < m < n$ and $n/m \in M$) has exactly one factorization. (a) If n has one factorization, $r_n = 1/k$ when $n = u^k$ for an atom u and $r_n = 0$ otherwise. (b) If n has $j \geq 2$ factorizations, $r_n = 1 - j + \varepsilon$, where $\varepsilon = \sum 1/k$ over the factorizations of the form u^k ; this is at most $-1/6$. Consequently M is free if and only if all $r_n \geq 0$, and if M is not free the first negative coefficient sits exactly at the least element with two factorizations. In a congruence monoid M_H with $H \neq G$ (§2.3) the value in (b) is at most $-1/2$, and infinitely many elements satisfy (b): for distinct primes p, r in a class $c \notin H$ and p', r' in $c^\wedge \{-1\}$, $n = pp'rr'$ has $r_n = -1$, or -2 if $c^2 \in H$ (for example $r_{\{3 \cdot 7 \cdot 11 \cdot 19\}} = -2$ in the Hilbert monoid $\{n \equiv 1 \pmod{4}\}$).

Proof. The entries of a tuple counted by $T_k(n)$, $k \geq 2$, are proper M -divisors of n and factor uniquely, so $T_k(n) = \sum_\Phi S_k(\Phi)$, where Φ runs over the factorizations of n and $S_k(\Phi)$ counts the ordered splittings of Φ into k nonempty parts. With a separate variable x_u for each atom u of Φ (atoms need not be multiplicatively independent: 4 and 8), splittings are counted by $\log \prod_u (1 - x_u)^{-1}$, so $\sum_k (-1)^{k+1} S_k(\Phi)/k$ is $1/k$ if Φ is k copies of one atom and 0 otherwise. This gives (a), and in (b) $r_n = 1 - j + \varepsilon$, since $T_1(n) = 1$ while there are j factorizations. The pure powers among them have distinct exponents ≥ 2 , so $r_n \leq -1/2$ if there is at most one, and $r_n \leq 1 - P + 1/2 + \dots + 1/(P + 1) \leq -1/6$ if there are $P \geq 2$. In M_H two pure-power factorizations $u^a = v^b$ would make u and v powers of one w with $[w] \in H$, against atomicity,

so $r_n \leq -1/2$. For $n = pp'rr'$ the factorizations are $\{pp', rr'\}$, $\{pr', rp'\}$, and $\{pr, p'r'\}$ when $c^2 \in H$, with $\varepsilon = 0$, and Dirichlet's theorem gives infinitely many such n . ■

Status. Verified: proved here. First proved in the programme (CEI Theorem 3.1 for the criterion, ETR1.5–1.7 for the bound $-1/6$) and derived independently in the audit; the hypothesis on proper M -divisors, the examples and the bound $-1/2$ were found in the audit (*new*). Novelty: not located in the literature (one search); elementary.

Corollary 2.5 (the first negative coefficient above $-1/2$). If $N \in M$ has two or more factorizations, every proper M -divisor of N has one, and $r_N > -1/2$, then N has exactly two factorizations $a^{\{j_1\}} = b^{\{j_2\}}$ with $\gcd(j_1, j_2) = 1$, $N = c^{\{j_1 j_2\}}$, and $r_N = -1 + 1/j_1 + 1/j_2$. Every value $-1 + 1/u + 1/v$ with $2 \leq u < v$ coprime occurs, at $N = c^{\{uv\}}$ in $\langle c^u, c^v \rangle$; so the possible first negative coefficients in $(-1/2, 0)$ are exactly $\{-1/2 + 1/v : v \geq 3 \text{ odd}\} \cup \{-5/12, -7/15\}$.

Proof. By Theorem 2.4(b), $r_N > -1/2$ forces $j = 2$ with both factorizations pure powers, $a^{\{j_1\}} = b^{\{j_2\}}$; $g = \gcd(j_1, j_2) > 1$ would make $a^{\{j_1/g\}} = b^{\{j_2/g\}}$ a proper M -divisor with two factorizations, so $g = 1$, and comparing prime exponents gives $a = c^{\{j_2\}}$, $b = c^{\{j_1\}}$; in $\langle c^u, c^v \rangle$ (whose generators are atoms since $u \nmid v$) the least exponent with two representations is uv ; and $1/u + 1/v > 1/2$ leaves $u = 2$ with v odd, or $u = 3$ with $v \in \{4, 5\}$. ■ *Status.* Verified: proved here and checked exactly on 286 generator sets; found in the audit (*new*); elementary, not searched.

Example 2.6. In $M = \{1\} \cup \{2^e : e \geq 2\}$, $64 = 4 \cdot 4 \cdot 4 = 8 \cdot 8$ and $r_{64} = -1/6$, while $D_M = (1 - x + x^2)/(1 - x)$, $x = 2^{-s}$, has all its zeros on $\operatorname{Re} s = 0$; so D_M has no zeros on its half-plane of absolute convergence $\operatorname{Re} s > 0$, like the Dirichlet series of every free monoid (ETR2), and M is not free. The monoid appears first in the programme's ETR1 (verified: a direct computation).

2.3 Congruence monoids and the even lattice sums

Fix $q \geq 1$ and a subgroup H of $G = (\mathbb{Z}/q)^\times$, and let $M_H = \{n \geq 1 : \gcd(n, q) = 1, n \bmod q \in H\}$; then $D_{\{M_H\}}(s) = [G : H]^{-1} \sum_{\chi|_H=1} L(s, \chi)$.

Proposition 2.7. The following are equivalent: (i) M_H is free; (ii) $H = G$; (iii) all $r_n \geq 0$; (iv) $D_{\{M_H\}}$ has no zeros in $\operatorname{Re} s > 1$. Moreover M_H is half-factorial if and only if $[G : H] \leq 2$.

Proof. (i) $\Leftrightarrow$ (iii) is Theorem 2.4, and M_G is free on the primes prime to q . If $H \neq G$, infinitely many primes lie outside H , so a class of order $d \geq 2$ in G/H contains primes $p \neq r$, and the atoms $A_j = p^{\{d-j\}} r^j$ satisfy $A_0 A_2 = A_1^2$; moreover $D_{\{M_H\}}$ then has components along two distinct primitive characters, so [SW] gives zeros in $\operatorname{Re} s > 1$ (such representations are unique; proof of Proposition 2.9). If all $r_n \geq 0$, then $\log D \leq D - 1$ coefficientwise, so $\log D$ converges absolutely on $\operatorname{Re} s > 1$ and $D = e^{\{\log D\}} \neq 0$ there. For $[G : H] = 2$ every factorization of n has length $a + b/2$, a and b counting the prime factors inside and outside H ; for $[G : H] \geq 3$, $(pp')^N = p^N p'^N$ or $(prs)^2 = p^2 \cdot r^2 \cdot s^2$ has factorizations of two lengths. ■

Status. Verified: proved here. (i) $\Leftrightarrow$ (ii) and the half-factoriality criterion are the standard criteria for Krull monoids with class group G/H (for $H = \{1\}$: [BC, Theorem 3.4], the Krull-monoid analogue of Carlitz's class-number-two theorem [Ca]); the direct proofs and the equivalences with (iii), (iv) were found in the audit (*new*), (iv) $\Rightarrow$ (ii) using [SW] as stated in its abstract.

Proposition 2.8 (the even sheets at $a = 1/q$). For $q \geq 3$, $Z_{\{1/q\}}(s) = q^s \sum_{n \equiv \pm 1 \pmod{q}} n^{-s} = q^s D_{\{M_{\{\pm 1\}}\}}(s)$. Hence the sheet $a = 1/q$ is Eulerian (the monoid $\{n \equiv \pm 1 \pmod{q}\}$ is free) if and only if $\varphi(q) \leq 2$, that is $q \in \{3, 4, 6\}$, if and only if its formal logarithm is nonnegative, if and only if $Z_{\{1/q\}}$ has no zeros with $\operatorname{Re} s > 1$; and exactly these q make $\{n \equiv 1 \pmod{q}\}$ half-factorial.

Proof. $\zeta(s, 1/q) = q^s \sum_{n \geq 0} (qn + 1)^{-s}$ and $\zeta(s, 1 - 1/q) = q^s \sum_{m \equiv -1 \pmod{q}} m^{-s}$; then apply Proposition 2.7 with $H = \{\pm 1\}$ and $H = \{1\}$. ■ *Status.* Verified: proved here; found in the audit (*new*). The packaging as a test on the sheets is not in the survey [BC]; not searched further.

Proposition 2.9 (zeros of the even sheets on both sides of the line). Let $\varphi(q) > 2$. There is $\eta > 0$ such that for all $1/2 < \sigma_1 < \sigma_2 < 1 + \eta$ the number of zeros of $Z_{\{1/q\}}$ with $\sigma_1 < \operatorname{Re} s < \sigma_2$ and $|\operatorname{Im} s| \leq T$ lies between $c_1 T$ and $c_2 T$ for all large T ($c_1, c_2 > 0$ depending on q, σ_1, σ_2), and the same holds, with another η , in the reflected strips $1 - \sigma_2 < \operatorname{Re} s < 1 - \sigma_1$. So $Z_{\{1/q\}}$ has infinitely many zeros with $\operatorname{Re} s > 1$, with $1/2 < \operatorname{Re} s < 1$, with $0 < \operatorname{Re} s < 1/2$ and with $\operatorname{Re} s < 0$.

Proof. On the right, $Z_{\{1/q\}} = q^s D_{\{M_{\{\pm 1\}}\}}$ has components along two distinct primitive characters, and [SW] applies. On the left, $Z_a(1 - s) = 2\Gamma(s)(2\pi)^{-s} \cos(\pi s/2) \Phi_a(s)$ with $\Phi_a(s) = 2\sum_{k \geq 1} \cos(2\pi k a) k^{-s}$, and at $a = 1/q$, expanding the cosines in even characters with Gauss sums gives $\Phi_{\{1/q\}} = \sum_{\psi} P_{\psi} L(s, \psi)$ over distinct primitive ψ with Dirichlet polynomials P_{ψ} . This representation is unique (at the least integer d_0 of the supports, the coefficients at $d_0 p$ for large primes p are $\sum_{\psi} b_{\{\psi, d_0\}} \psi(p)$, and distinct primitive characters are independent on the primes); the ζ -component is nonzero (its coefficient at q^{-s} is $2\sum_{e|q \text{ squarefree}} 1/\varphi(e)$), and another is nonzero, since otherwise $\cos(2\pi p/q) = \cos(2\pi/q)$ for all large primes p , forcing $\varphi(q) \leq 2$. So [SW] gives zeros s_0 of $\Phi_{\{1/q\}}$ in the strips, and each $1 - s_0$ is a zero of $Z_{\{1/q\}}$, the factor $2\Gamma(s_0)(2\pi)^{-s_0} \cos(\pi s_0/2)$ being finite and nonzero. ■

Status. Verified: proved here, using [SW] as stated in its abstract (its printed statement was not checked); found in the audit (*new*); check Z8 finds zeros of $Z_{\{1/8\}}$ and $Z_{\{1/12\}}$ in $0 < \operatorname{Re} s < 1/2$. Numerically, $Z_{\{1/5\}}$ (whose monoid first has two factorizations at $36 = 4 \cdot 9 = 6 \cdot 6$, with $r_{36} = -1/2$) has 24 zeros with $\operatorname{Re} s > 0.505$ below height 150, the lowest at $0.54307 + 15.70405i$, and zeros to the left of the line such as $-0.158 + 6.614i$.

2.4 Shifted zeta products and tensor Euler functions

For finite exponent data, a finitely supported measure μ on $\mathbb{C}$ with positive integer masses, the programme's tensor Euler function of degree d is $\mathcal{L}_d = \prod_p \det(1 - p^{-s} W_p^{\{\otimes d\}})^{-1} = \prod_{\sigma} \zeta(s - \sigma)^{\mu^{\{*d\}}(\{\sigma\})}$ (ETR4-ETR5).

Lemma 2.10 (shifted zeta functions are multiplicatively independent). Let $\sigma_1, \dots, \sigma_{\ell} \in \mathbb{C}$ be distinct and $C_j \in \mathbb{C}$. If $\sum_j C_j (\zeta/\zeta)(s - \sigma_j) \equiv 0$, then all $C_j = 0$; in particular $\prod_j \zeta(s - \sigma_j)^{C_j} \equiv 1$ with $C_j \in \mathbb{Z}$ only if all $C_j = 0$. The same holds for $L(s - \sigma_j, \chi)$, χ any Dirichlet character.

Proof. The Euler logarithm $\sum_j C_j \log \zeta(s - \sigma_j) = \sum_p \sum_r (1/r) (\sum_j C_j p^{\{r\sigma_j\}}) p^{-rs}$ vanishes identically (its derivative is the given sum, and it tends to 0 as $s \rightarrow +\infty$), so $\sum_j C_j p^{\{r\sigma_j\}} = 0$ for all p and $r \geq 1$ (for L , after dividing by $\chi(p)^r$, $p \nmid q$). Two points with $p^{\{\sigma_i\}} = p^{\{\sigma_j\}}$ cannot collide at two primes, so for some prime the $x_j = p^{\{\sigma_j\}}$ are distinct, and the system for $r = 1, \dots, \ell$ has determinant $\prod_j x_j \prod_{\{i < j\}} (x_j - x_i) \neq 0$. ■

Lemma 2.11 (uniqueness of convolution roots). If finitely supported complex measures satisfy $\mu^{\{*d\}} = \nu^{\{*d\}}$, then $\mu = \eta\nu$ for a d -th root of unity η , and $\mu = \nu$ if the total masses are positive reals.

Proof. The exponential transforms satisfy $A^d = B^d$; on the connected set $\{B \neq 0\}$, A/B is a continuous d -th root of unity, hence constant, and distinct exponentials are linearly independent. ■

Corollary 2.12. For finite exponent data a single $\mathcal{L}_d$ determines μ (Lemma 2.10 gives $\mu^{\{*d\}}$, and Lemma 2.11 gives μ).

Status of 2.10-2.12. Verified: proved here; proved in the programme at ETR8, ETR9 and ETR4-ETR9 for integer exponents and positive masses (the complex, L -function and root-of-unity forms found in the audit, *new*). For conjugation-stable data the real pole of $\mathcal{L}_{\{2k\}}$ at $1 + 2kB$ has order $h_{\{2k\}}$, with $h_{\{2k\}}^{\{1/2k\}}$ increasing to the dimension of the maximal face (ETR5-ETR7): Deligne's §1.5 mechanism, as the programme credits. Lemma 2.10 is necessarily global (§8, N-Eu).

3. Meyer's quotient, its jets and its receivers

Every quotient of $\mathcal{B}$ by a closed subspace is a Fréchet space with the quotient seminorms $\bar{b}_{\{A, M\}}$ [Ru, Theorem 1.41].

3.1 The strip space and multiplication by s

Lemma 3.1. For all $A, M \geq 0$ the unit ball of $b_{\{A+1, M+1\}}$ is precompact for $b_{\{A, M\}}$. So $\mathcal{B}$ and every quotient $\mathcal{B}/I$ by a closed subspace are Fréchet-Montel, hence reflexive, and $(\mathcal{B}/I)'_{\beta} \cong I^{\perp} \subset \mathcal{B}'_{\beta}$ topologically.

Proof. Cauchy's estimate makes the ball equicontinuous on $|\operatorname{Re} s| \leq A + 1/2$, the extra factor $(1 + |t|)^{-1}$ makes $|t| \geq T$ uniformly small, and Arzelà-Ascoli on the remaining rectangle gives finite nets; the rest is the permanence of Fréchet-Schwartz spaces under closed quotients [Ja; SchW, Ch. IV]. ■ *Status.* Verified: proved here; found in the audit (*new*). The programme obtains these properties through the summation image (SDT).

Proposition 3.2 (multiplication by s on $\mathcal{B}/I_D$). Let $0 \neq \Phi \in \mathcal{B}$ vanish to order at least D . (a) For $\lambda \notin \operatorname{supp} D$, $\lambda - L$ is invertible on $\mathcal{B}/I_D$, with inverse induced by $R_\lambda F = (F - \Phi_\lambda F(\lambda)/\Phi_\lambda(\lambda))/(\lambda - s)$, $\Phi_\lambda = \Phi/(s - \lambda)^{\{\operatorname{ord}_\lambda \Phi\}}$. (b) The spectrum of L is $\operatorname{supp} D$. (c) If ρ has multiplicity m in D , then $\ker(L - \rho)^m = \ker(L - \rho)^{m+1}$ has dimension m , j_ρ identifies it with $\mathbb{C}[T]/(T^m)$, and $L - \rho$ acts on it as multiplication by T : one Jordan block of length m .

Proof. Division by $(s - \lambda)^k$ at a zero of order $\geq k$ preserves $\mathcal{B}$ (maximum principle near λ), so $\Phi_\lambda \in I_D$ and R_λ inverts $\lambda - s$ modulo I_D . With $m' = \operatorname{ord}_\rho \Phi$, $U = \Phi/(s - \rho)^{\{m' - m + 1\}}$ is not in I_D while $(s - \rho)U$ is. A class lies in $\ker(L - \rho)^k$ exactly when it vanishes to full order off ρ and to order $m - k$ at ρ , and $U_\rho P(s - \rho)$, $U_\rho = \Phi/(s - \rho)^{\{m'\}}$, realizes every jet at ρ . ■ *Status.* Verified: proved here; proved in the programme for the zeta divisor at RZ3-RZ7; the general-divisor form was found in the audit (*new*).

Lemma 3.3 (no equivariant maps across disjoint divisors). Let $D \leq \operatorname{div} F_D$ for some $0 \neq F_D \in \mathcal{B}$, and let D'' be the part of a divisor D' off $\operatorname{supp} D$. Every linear $H : \mathcal{B}/I_D \rightarrow \mathcal{B}/I_{D'}$ with $HL = LH$ takes values in $I_{D''}/I_{D'}$; in particular $H = 0$ if the supports are disjoint. Continuity is not needed.

Proof. For $\lambda \in \operatorname{supp} D' \setminus \operatorname{supp} D$ of multiplicity m , $(L - \lambda)^m$ is onto $\mathcal{B}/I_D$, so every $H(y) = (L - \lambda)^m H(x)$ has zero m -jet at λ . ■ *Status.* Verified: proved here; found in the audit (*new*); it strengthens the programme's GSL6.

Lemma 3.4 (uniqueness of exponential sums). Let $0 \neq G \in \mathcal{B}$ and let Z be a set of zeros of G in a vertical strip, $m_\rho = \operatorname{ord}_\rho G$. If $\sum_{\rho \in Z} c_{\rho,k} x^k e^{\{ \rho x \}} = 0$ for all real x , with $0 \leq k < m_\rho$ and $\sum |c_{\rho,k}| k! R^{-k} < \infty$ for some $R > 0$, then all $c_{\rho,k} = 0$.

Proof. Pairing with the half-Mellin representation $F(s) = (1/2) \int_{\mathbb{R}} f(v) e^{\{sv\}} dv$ of $F \in \mathcal{B}$ (NJS4.1) gives $\sum c_{\rho,k} F^{\{k\}}(\rho) = 0$ for all $F \in \mathcal{B}$; test with $G(s)(s - \rho_0)^{-m} P(s - \rho_0)$, P a polynomial. ■ *Status.* Verified: proved here; in the programme for the Gaussian-weighted zeta divisor at HBW8.9-8.12; the general form was found in the audit (*new*).

3.2 The exact summation image

Let S be the space of even Schwartz functions f with $f(0) = 0$ and $\int f = 0$, $\Sigma f(u) = 2 \sum_{n \geq 1} f(nu)$, and $\mathcal{M}_0 b(s) = \int_0^\infty b(u) u^{s-1} du$.

Proposition 3.5 (the exact summation image). $f \mapsto \mathcal{M}_0 \Sigma f$, which equals $2\zeta(s) \int_0^\infty f(v) v^{s-1} dv$ on $\operatorname{Re} s > 1$, is a topological isomorphism of S onto I_ζ , with inverse $F \mapsto f_F(x) = (2\pi)^{-1} \int_{\mathbb{R}} F(2 + it) f_F(x) dt = (1/2) \sum_{n \geq 1} \mu(n) (\mathcal{M}_0^{-1} F)(nx)$; moreover $f_F^{\{2r\}}(0)/(2r)! = F(-2r)/(2\zeta'(-2r))$ for $r \geq 1$, $\int_0^\infty f_F dx/x = -F(0)$ and $\int_0^\infty f_F \log x dx = F(1)/2$. In particular the summation image is closed.

Outline. Poisson summation controls $u \rightarrow 0$, and Müntz's formula [Ti, §2.11] with the Euler product gives injectivity. For $F \in I_\zeta$, $F/(2\zeta)$ has simple poles only at $-2r$; the division estimate $|F/F_*| \leq C \exp(C(|t| + 2)^{3/2} \log(|t| + 2))$ (S3, OMS2.8) is removed by a factor $e^{\{\varepsilon s^2\}}$ and the maximum principle, and shifting the inverse Mellin contour to $\operatorname{Re} s = -2N - 1$ picks up the residues $F(-2r)/(2\zeta'(-2r)) x^{\{2r\}}$.

Status. Verified: proved in the programme at SSI1-SSI7 and OMS2-OMS4, and checked in the audit, including the Taylor coefficients and endpoint identities on the programme's source vector. On the programme's account (OMS0) this is Meyer's range theorem [Me] made explicit for $\mathcal{B}$; the comparison with Meyer's paper is not verified here. Consequently the closure of Connes' summation map in the rapid-decay Fréchet space (the ideal of zero-jets, S5 of the second attempt) is attained by the image itself, and $Q = \mathcal{A}/\mathcal{E}S \cong \mathcal{B}/I_\zeta$.

3.3 The jet map and the primary decomposition

The programme's section $s_\rho(P) = [U_\rho \cdot (Pc_\rho)(s - \rho)]$, $U_\rho = F_*/(s - \rho)^{\{m_\rho\}}$, c_ρ the inverse of the jet of U_ρ , satisfies $j_{ns_\rho} = \delta_{\{\eta\rho\}}$; so the $\Pi_\rho = s_\rho j_\rho$ are orthogonal commuting idempotents, and finite sums of sections realize every finitely supported jet tuple (OMS6). Let Λ_m be the Fréchet space of tuples $(P_\rho) \in \prod A_\rho$ with $\sup_\rho (1 + |\gamma|)^M \|P_\rho\| < \infty$ for all M , $\|\cdot\|$ the largest coefficient.

Lemma 3.6 (jet decay). For $F \in \mathcal{B}$, $|\beta| \leq 3/2$ and $r, M \geq 0$, $(1 + |\gamma|)^M |F^{\{r\}}(\rho)/r!| \leq 2^r ((1 + |\gamma|)/(1/2 + |\gamma|))^M b_{\{2,M\}}(F) \leq 2^{\{r+M\}} b_{\{2,M\}}(F)$, and at the nontrivial zeros the middle factor is at most 1.0342^M (Cauchy's formula on $|z - \rho| = 1/2$, and $\gamma_1 = 14.1347\dots$). *Status.* Verified: proved here; found in the audit (*new*), an explicit form of OMS5.6.

Corollary 3.7. The jet image $J(Q)$, $J = (j_\rho)_\rho$, lies in Λ_m and is dense there; it is a proper dense subspace of the product, the quotient topology of Q is strictly finer than the product topology on $J(Q)$, and Q is not the product of its primary blocks.

Proof. Lemma 3.6 and $m_\rho = O(\log |\gamma|)$ give $J(Q) \subset \Lambda_m$, and the finitely supported tuples, realized by sections, are dense in Λ_m and in the product; a class with $F(\rho) = 1$ at every zero would give $1 + |\gamma| \leq 2b_{\{2,1\}}(F)$ for all ordinates; and equal topologies would make $J(Q)$ complete, hence the whole product. ■ *Status.* Verified: proved in the programme at OMS5 and checked in the audit; the density in Λ_m was found in the audit (*new*).

We use (D1) the number of zeros with $|\gamma - T| < 1$ is $O(\log T)$, (D2) $\zeta'/\zeta(s) = \sum_{|t-\gamma|<1} (s - \rho)^{-1} + O(\log t)$ for $-1 \leq \sigma \leq 2$, $t \geq 2$ [Ti, Theorems 9.2, 9.6(A)], and **Lemma 3.8** [Ti, Theorem 9.7]: there are A, T_0 with a good height t in every $[T, T + 1]$, $T \geq T_0$, where $|\zeta(\sigma \pm it)| \geq t^{-A}$ for $-1 \leq \sigma \leq 2$ (verified: Titchmarsh's proof; the statements were checked on the printed pages). Let $c_{\{\rho,k\}}$ ($k < m = m_\rho$) be the Taylor coefficients at ρ of $(s - \rho)^m/\zeta(s)$, so that $c_{\{\rho,0\}} = m!/\zeta^{\{m\}}(\rho)$.

Theorem 3.9 (when the primary decomposition converges). The following are equivalent.

- (a) For some enumeration (ρ_n) of the distinct zeros, $\sum_{k \leq n} \Pi_{\{\rho_k\}} x$ converges for every $x \in Q$ (the limit is then x).
- (a') For every $x \in Q$ the set $\{\Pi_{\rho x}\}_\rho$ is bounded.
- (b) $\sum_\rho \Pi_{\rho x}$ converges absolutely for every $x \in Q$.
- (b') For every x , every continuous seminorm q and every N , $q(\Pi_{\rho x}) = O((1 + |\gamma|)^{-N})$.
- (c) There are C, K with $|c_{\{\rho,k\}}| \leq C(1 + |\gamma|)^K$ for all ρ and $k < m_\rho$: the principal parts of $1/\zeta$ at the zeros are polynomially bounded.
- (d) $J : Q \rightarrow \Lambda_m$ is an isomorphism of Fréchet spaces.

For simple zeros (c) reads $|\zeta'(\rho)| \geq c(1 + |\gamma|)^{-K}$, and in every case (a') implies $|\zeta^{\{m_\rho\}}(\rho)/m_\rho!| \geq c(1 + |\gamma|)^{-K}$. The theorem carries over to the zeros of a primitive Dirichlet L -function, with $1/L(s, \chi)$ in (c), $L(s, \chi)$ in place of $(s - 1)\zeta(s)$ below, and (D1), (D2) and the good heights with $\log q(|t| + 2)$.

Proof. (a) $\Rightarrow$ (a') and (b) $\Rightarrow$ (a) are immediate, and a limit y of the partial sums has the jets of x , so $y = x$.

(a') $\Rightarrow$ (c). By Banach–Steinhaus [Ru, Theorem 2.6], $\bar{b}_{\{2,0\}}(\Pi_{\rho x}) \leq C_0 \bar{b}_{\{A_1, M_1\}}(x)$ for some A_1, M_1, C_0 . The Gaussian $E_\rho(s) = e^{\{s-\rho\}^2}$ (CFP6.1) has $b_{\{A_1, M_1\}}(E_\rho) \leq C_1(1 + |\gamma|)^{M_1}$, so $\Pi_\rho[E_\rho]$ has a representative F_ρ with $b_{\{2,0\}}(F_\rho) \leq 2C_0C_1(1 + |\gamma|)^{M_1}$, jet $e^{\{T^2\}}$ at ρ , and full-order vanishing at the other zeros. For $\gamma \geq T_0 + 2$ take good heights $T_1 \in [\gamma - 2, \gamma - 1]$, $T_2 \in [\gamma + 1, \gamma + 2]$ and $R = [-1, 2] \times [T_1, T_2]$. $H = F_\rho(s - \rho)^m/\zeta$ is holomorphic near R (F_ρ cancels the other zeros in R , and $\zeta \neq 0$ on ∂R), and on ∂R , $|1/\zeta| \leq \pi^2/6$ on $\sigma = 2$, $|1/\zeta| \leq T_2^A$ on the horizontal sides, and $|\zeta(-1 + it)| = |\chi(-1 + it)| |\zeta(2 - it)| \geq 3/\pi^4$ on $\sigma = -1$, from $|\chi(-1 + it)|^2 = \pi^{\{ -3 \}} y \coth(\pi y) (1/4 + y^2) \geq 1/(4\pi^4)$, $y = t/2$. So $|H| \leq B_\rho := 2C_0C_1(1 + |\gamma|)^{M_1} 3^m \max(\pi^2/6, \pi^4/3, (\gamma + 2)^A)$ on R . The jet of H at ρ is $e^{\{T^2\}} \sum_{k \leq m} c_{\{\rho,k\}} T^k$, and Cauchy's estimate on $|s - \rho| = 1/2$ gives $|c_{\{\rho,k\}}| \leq 2^m B_\rho$, which is polynomial in $|\gamma|$ because $m \leq C \log(2 + |\gamma|)$ by (D1).

(c) $\Rightarrow$ (d) and (b). $V_\rho = (s - 1)\zeta(s)/(s - \rho)^m$ is entire, and its reciprocal jet $v_\rho^{\{-1\}} = (\sum_{k \leq m} c_{\{\rho,k\}} T^k)(\rho - 1 + T)^{-1}$ has coefficients at most $mC(1 + |\gamma|)^K$. With $\lambda = m + 1$ and $q_{\{\rho,k\}} =$

$[T^k e^{-\lambda T^2} v_\rho^{-1}]_{\{<m\}}$, $G_{\{\rho,k\}} = e^{\{\lambda(s-\rho)^2\}} V_{\rho,q_{\{\rho,k\}}}(s-\rho)$ is entire, has jet T^k at ρ , vanishes to full order at the other zeros, and, since $|e^{\{\lambda(s-\rho)^2\}}| \leq e^{\{(m+1)(A+1)^2\}} e^{\{-(m+1)(t-y)^2\}}$ and $(s-1)\zeta(s)$ grows polynomially on strips [Ti, Ch. V], satisfies $b_{\{A,M\}}(G_{\{\rho,k\}}) \leq C_{\{A,M\}}(2 + |y|)^{N_{\{A,M\}}}$. As $\Sigma_{\rho m_\rho}(2 + |y|)^{-2} < \infty$, $F_P = \Sigma_{\{\rho,k\}} P_{\{\rho,k\}} G_{\{\rho,k\}}$ converges absolutely in $\mathcal{B}$ for $P \in \Lambda_m$, with $J[F_P] = P$; so J is an isomorphism, and $\Sigma_{kx_{\{\rho,k\}}}[G_{\{\rho,k\}}] = \Pi_{\rho x}$ gives $x = \Sigma_{\rho} \Pi_{\rho x}$ absolutely.

(d) $\Rightarrow$ (b') $\Rightarrow$ (b). $q(\Pi_{\rho x}) = q(J^{-1}(\Sigma_{kx_{\{\rho,k\}}} \delta_{\{\rho,k\}})) \leq C(1 + |y|)^{-N} p_{\{M+N\}}(Jx)$, and (b') with (D1) gives absolute convergence. ■

Status. Verified: proved here; found in the audit (*new*), in the form for all multiplicities as sharpened in a referee pass; checks F1-F9, G1-G3. Novelty: not found in the programme or in the literature searched [BT, AKN, BRS]; it is an instance for Meyer's quotient of the classical link between lower bounds at the zeros and interpolation. The $L(s, \chi)$ form is an outline, and its standard inputs are not verified here.

Remark 3.10. Whether (c) holds is open. A bound $|\zeta'(\rho)| \geq c(1 + |y|)^{-K}$ at every zero would imply that every zero is simple, which is open; unconditionally at least a proportion 0.4075 of the zeros are simple (0.407511 for simple zeros on the line [PRZZ]; earlier 0.4058 [BCY], as cited in [BHB, §1]). Under RH one expects $|\zeta'(\rho)|^{-1} \gg |y|^{\{1/3-\varepsilon\}}$ for infinitely many zeros [MN], so $K \geq 1/3$ would be needed. For multiple zeros, whether the leading coefficients suffice is not settled; in the spaces A_p they do [BT, Theorem 4]. The programme itself uses only finite sums of primary projectors (OMS6, SCL7, GEX2).

3.4 Dense families of test functions

For $t > 0$ put $g_t(s) = e^{\{ts^2\}}$, $\mathcal{N}_S = \text{span}\{g_t(s)(n - n^{\{1-s\}})/s : n \in S\}$ for $S \subset \{2, 3, \dots\}$, and $v_S(x) = \#\{n \in S : \log n \leq x\}$.

Proposition 3.11 (Nyman-Beurling in the Fréchet topology). If (C1) $\limsup_R R^{-1} \Sigma_{\{n \in S, \log n < R\}} (1/\log n - \log n/R^2) > 1/(3\pi t)$, or (C2) $\limsup_X (\log X)^{-1} \Sigma_{\{n \in S, \log n < X\}} ((\log n)^{-2} - (\log n)^2/X^4) > 1/(4\pi t)$, then $\mathcal{N}_S$ is dense in $\mathcal{B}$ and $\xi \mathcal{N}_S$ is dense in I_ζ . (C1) holds for every t if $\limsup v_S(R)/R^2 = \infty$ (for example $S = \{2, 3, \dots\}$, or the d -th powers). If $v_S(x) \sim \kappa x^2$, (C1) holds exactly when $t > 1/(4\pi\kappa)$ and (C2) exactly when $t > 1/(8\pi\kappa)$; for $S = \{2^a 3^b\}$, $\kappa = 1/(2 \log 2 \log 3)$, so this family is dense for every $t > (\log 2)(\log 3)/(4\pi) = 0.06060\dots$ ((C1) alone gives $0.12120\dots$). Density for one t implies density for every larger t , and the closure of $\xi \mathcal{B}$ is I_ζ although $\xi \mathcal{B} \subsetneq I_\zeta$.

Proof. 1. Let $\Lambda \in \mathcal{B}'$ vanish on $\mathcal{N}_S$, $|\Lambda(F)| \leq C b_{\{A,M\}}(F)$. With $K_w = g_t(s)(e^w - e^{\{(1-s)w\}})/s$, a $\mathcal{B}$ -valued power series in w , $L(w) = \Lambda(K_w)$ is entire, $L(0) = 0$ and $L(\log n) = 0$ for $n \in S$.

2. For $|\sigma| \leq A$, $s = \sigma + iy$ and $w = u + iv$, the bound $-ty^2 + |y||v| \leq v^2/(4t)$ gives $\log|L(u + iv)| \leq v^2/(4t) + (1 + A)|u| + M \log(2 + |v|) + C'$, with $1/(4t)$ independent of Λ .
3. If $L \not\equiv 0$, Carleman's formula in the right half-plane [TiF, §3.7] bounds $\Sigma_j (1/r_j - r_j/R^2) \cos \varphi_j$ over the zeros $r_j e^{i\varphi_j}$ by the arc term, at most $(R/(4\pi t)) \int \sin^2 \varphi \cos \varphi d\varphi = R/(6\pi t)$, the axis term, at most $R/(6\pi t) + O(1)$, and $O(\log R)$. The zeros $\log n$ lie on the positive axis and the other zeros add nonnegative terms, so $\Sigma_{\{\log n < R\}} (1/\log n - \log n/R^2) \leq R/(3\pi t) + O(\log R)$, contradicting (C1). 3'. Apply the same formula to $L(z^{\{1/2\}})$ on $|z| \leq X^2$, whose zeros include $(\log n)^2$: by step 2 the arc contributes at most $(1 - \pi/4)/(4\pi t) + o(1)$ and the axis, where $v^2 = y/2$, at most $(\log X)/(4\pi t) + O(1)$, so $\Sigma_{\{\log n < X\}} ((\log n)^{-2} - (\log n)^2 X^{-4}) \leq (\log X)/(4\pi t) + O(1)$, contradicting (C2).
4. So $L \equiv 0$; since $(\partial_w - 1)K_w = g_t e^{\{(1-s)w\}}$, $\Lambda(g_t e^{\{vs\}}) = 0$ for all real v , and the half-Mellin representation of $G \in \mathcal{B}$ (NJS4.1) gives $\Lambda(g_t G) = 0$. Finally $g_t \mathcal{B}$ is dense: with $h_N = e^{\{ts^2\}} P_N(-ts^2)$, P_N the Taylor polynomial of $\exp$, $Fh_N \in g_t \mathcal{B}$, $|h_N| \leq e^{\{2t\sigma^2\}}$ and $h_N \rightarrow 1$ locally uniformly, so $b_{\{A,M\}}(Fh_N - F) \rightarrow 0$.
5. If Λ vanishes on $\xi \mathcal{N}_S$, then $F \mapsto \Lambda(\xi F)$ vanishes on $\mathcal{N}_S$, so $\Lambda = 0$ on $\xi \mathcal{B}$, which is dense in I_ζ (NCI2).
6. Partial summation gives $\Sigma_{\{\log n < R\}} (1/\log n - \log n/R^2) = (4/3)\kappa R + o(R)$ and $\Sigma_{\{\log n < X\}} ((\log n)^{-2} - (\log n)^2 X^{-4}) = 2\kappa \log X + o(\log X)$ when $v_S(x) \sim \kappa x^2$, hence the thresholds; and $\mathcal{N}_S^{\{(t)\}} = g_{\{t-t'\}} \mathcal{N}_S^{\{(t')\}}$ with $g_{\{t-t'\}} \mathcal{B}$ dense gives monotonicity in t . ■

Status. Verified: proved here. Proved in the programme for $S = \{2, 3, \dots\}$ at NCI2-NCI8 and RSS1-RSS2; the sparse version, conditions (C1) and (C2) and the complete proof were found in the audit (*new*; check Z7). Carleman's formula is used as stated in [TiF, §3.7] (not re-read; its half-plane and angle forms were checked numerically on test functions). The corresponding statement in Noor's Hardy-space setting is equivalent to RH [No]; here it is unconditional, because the topology is different.

Proposition 3.12 (an infinite set of degrees need not suffice). If $S \subset \{a^k : k \geq 1\} \cup F$ with an integer $a \geq 2$ and F finite, the closures of $\mathcal{N}_S$ in $\mathcal{B}$ and of $\xi\mathcal{N}_S$ in I_ζ have infinite codimension; if $F = \emptyset$, the closure of $\mathcal{N}_S$ lies in I_D with $D = \Sigma_{\{j \neq 0\}}[2\pi i j / \log a]$.

Proof. Every generator with $n = a^k$ vanishes at $s_j = 2\pi i j / \log a$, $j \neq 0$; point evaluations are linearly independent on $\mathcal{B}$, so for each J the combinations of $J + |F|$ evaluations killing the generators with $n \in F$ form a space of dimension $\geq J$. In I_ζ the same holds, since $\xi(s_j) = \xi(1 - s_j) \neq 0$. ■ *Status.* Verified: proved here; found in the audit (*new*).

If Φ is entire, $\Phi(0) = 1$ and $g_t(1 - \Phi)/s \in \mathcal{B}$, the family $\{g_t(n^{1-s} - n\Phi)/s : n \geq 1\}$ is dense in $\mathcal{B}$, its span containing $g_t(1 - \Phi)/s$ and hence $\mathcal{N}_{\{2,3,\dots\}}$ (verified; found in the audit). The statements on tails of Noor's family with the programme's correction $C = 8F_*$ (total in $\mathcal{B}$; annihilator 0 or $\mathbb{C}\cdot\text{ev}_1$ according as $C(1) \neq 0$ or $C(1) = 0$) were found in the audit and are not verified here.

3.5 Positive forms and receivers

Lemma 3.13 (transfer-compatible forms). On $H = \ell^2(\mathbb{Z}, m)$, $\|x\|^2 = \sum m_\rho |x_\rho|^2$, let $(T_{nx})_\rho = n^\rho x_\rho$ and $U_n = nT_{\{1/n\}}$, and let B be a bounded Hermitian form with $B(T_{nx}, T_{ny}) = nB(x, y)$ for $n = 2$ and $n = 3$. Then $B(x, y) = \sum_\rho b_\rho m_\rho x_\rho \text{conj}(y_{\rho^\#})$ with $|b_\rho| \leq \|B\|$ and $b_{\rho^\#} = \text{conj}(b_\rho)$, and every such form satisfies the relation for all real $n > 0$. B is positive semidefinite if and only if $b_\rho = 0$ at every off-line zero and $b_\rho \geq 0$ on the line; in particular the bounded positive forms with $B(T_{nx}, y) = B(x, U_{ny})$ for all integers $n \geq 1$ are exactly $\sum_{\{\text{Re } \rho = 1/2\}} m_\rho c_\rho x_\rho \text{conj}(y_\rho)$, $0 \leq c_\rho \leq C$. The pair $\{2, 3\}$ may be any $n_1, n_2 > 1$ with $\log n_1 / \log n_2$ irrational; a single n allows further entries at (ρ, η) with $\eta - \rho^\# \in (2\pi i / \log n)\mathbb{Z}$.

Proof. On coordinate vectors the relation reads $(n^{\rho+\eta-1} - 1)B(\varepsilon_\rho, \varepsilon_\eta) = 0$, and $(2\pi i / \log 2)\mathbb{Z} \cap (2\pi i / \log 3)\mathbb{Z} = \{0\}$, so nonzero entries have $\eta = \rho^\#$; boundedness and symmetry give the rest, and $\rho + \text{conj}(\rho^\#) = 1$ the converse. An off-line pair contributes $2m_\rho \text{Re}(b_\rho x_\rho \text{conj}(x_{\rho^\#}))$, with eigenvalues $\pm m_\rho |b_\rho|$, so positivity forces $b_\rho = 0$ there and $b_\rho \geq 0$ on the line; $T_{n^{-1}} = T_{\{1/n\}}$ makes the adjoint relation equivalent. ■

Status. Verified: proved here; proved in the programme for positive forms and all n at PTQ4; the classification of all bounded Hermitian forms from $n = 2, 3$ was found in the audit (*new*). With $b \equiv 1$ the diagonal of the form is Weil's pairing Ω : two multiplicatively independent integers already force this shape, and positivity is exactly what removes the off-line blocks.

The following are proved in the programme at the labels given and checked in the audit (verified), unless an item says otherwise.

- **Positive transfer forms** on $\mathcal{B}/I_\zeta$ are exactly $\sum_{\{\text{Re } \rho = 1/2\}} m_\rho c_\rho F(\rho) \text{conj}(G(\rho))$ with polynomially bounded $c_\rho \geq 0$ (CFP); before the quotient they are $\int F(1/2 + i\lambda) \text{conj}(G(1/2 + i\lambda)) d\mu(\lambda)$, μ positive and tempered [VI], and they descend exactly when $\text{supp } \mu$ lies in the ordinates of critical zeros (SPF0-SPF10). For every positive μ with polynomially bounded mass on unit intervals, the closure of I_ζ in $L^2(\mu)$ under $F \mapsto F(1/2 + i\lambda)$ is $L^2(\mathbb{R} \setminus \Gamma, \mu)$, Γ the ordinates of critical zeros (SMC1-SMC4, SMC9).
- **Connes against Meyer.** The map from Q to the inverse limit of Connes' weighted Hilbert quotients $[Co]$ has kernel exactly the classes vanishing on all critical-line jets, so it is injective if and only if RH holds; the loss happens in the Hausdorff quotient (WHR5-WHR10). The prime dilations there have spectrum on $|z| = \sqrt{p}$ (WHR8), and the analogous quotients on the lines $\text{Re } s = \sigma$, $0 < \sigma < 1$, realize exactly the zeros on those lines and are jointly faithful on Q (VWR2-VWR7).
- **Spectral synthesis.** Finite-rank Riesz-Gaussian operators converge to the identity on Q ; closed multiplier submodules are the jet-order submodules; the line and off-line ideals have dense sum (GSP2-GSP6, CTS1-CTS4; read for correctness, not re-derived).

- **Global residue duality.** $B_{\zeta}(F, G) = (2\pi i)^{-1} \left(\int_{\text{Re } s=2} - \int_{\text{Re } s=-1} \right) F(s)G(1-s)/\zeta(s)ds$ satisfies $|B_{\zeta}(F, G)| \leq \zeta(2)(\pi^{-1} + 8\pi/5)b_{\{2,1\}}(F)b_{\{2,1\}}(G) \approx 8.79 b_{\{2,1\}}(F)b_{\{2,1\}}(G)$, descends to Q , is nondegenerate with local determinant $(m!/\zeta^{\wedge\{m\}}(\rho))^m$, satisfies $B(T_{\text{ax}}, T_{\text{ay}}) = aB(x, y)$ and $B(P(L)x, y) = B(x, P(1-L)y)$, and is not positive (GZR).
- **The source-pairing identity.** If $b_0 \in \mathcal{A}$ has Mellin transform vanishing at ρ and $(L - \rho)b_{\rho} = b_0$ with $L = -u\partial_u$, then $(\text{Re } \rho - 1/2)\|b_{\rho}\|_{L^2(du)}^2 = -\text{Re}\langle b_{\rho}, b_0 \rangle$, since $\text{Re}\langle Lb, b \rangle = \|b\|^2/2$ (OPD4.2; proved here); it restates RH at ρ as $\text{Re}\langle b_{\rho}, b_0 \rangle = 0$, and reading it as a Hilbert-Pólya mechanism is an interpretation.
- **Harmonic defect and divisor potential.** The Poisson-swept distribution, with $H(b_{\dagger}) = 2\sum m_{\rho}|\text{Re } \rho - 1/2|\|b_{\rho}\|^2$, descends to Q if and only if there are no off-line zeros (HSW5-HSW6B, OPD5); the compressed translation trace tends to $\sum m e^{-|\beta-1/2||t|}e^{i\gamma t}$ (FGR0-FGR6); and $\mathcal{E}_r(t) = \sum_{\rho} m_{\rho}(r^{\sigma} - r^{1-\sigma})^2 e^{2t(\sigma^2 - \gamma^2)}$, zero if and only if RH holds, equals an area integral of $(1/2)\log|\zeta(s)\zeta(1-\bar{s})|$ against the Laplacian of an explicit test function plus $(r-1)^2(e^{2t} + 1)/2$, for a cutoff supported in $(-1/2, 3/2)$ (OZD0-OZD7; the Littlewood-lemma family [BSY, SBM]).

Each of these objects is exact. The positive ones are supported on the critical zeros; those that involve the off-line zeros vanish, or are injective, exactly under RH; and B_{ζ} is nondegenerate unconditionally and not positive. Deciding the location of the zeros through any of them therefore requires an input that forces the off-line part to vanish (§8.2).

4. The two lines

The author proposed a second line at $\text{Re } s = -1/2$ that mirrors the critical line, built by shifting the whole line as in the Hurwitz flow, and asked for the anomaly, chirality and holonomy language of that picture to be made exact. This section gives the exact objects; the matching of the author's terms with them is a proposal (§9.8). No result assumes RH, and Proposition 4.10 states its GRH hypotheses. Note $s^{\#} = T_1(A(s))$. The results were found in the audit (*new*) unless a status line says otherwise.

4.1 The first Hurwitz jet and the mirror set

Theorem 4.1 (the first Hurwitz jet). Let $a > 0$. (a) $\partial_a \zeta(s, a) = -s\zeta(s+1, a)$ for $s \neq 1$ (at $s = 0$ the right side is its holomorphic extension), and $\partial_a^m \zeta(s, a) = (-1)^m m! \zeta(s+m, a)$, both sides entire in s . (b) $s\zeta(s+1, a)$ is entire and equals 1 at $s = 0$. (c) The zero divisor of the first jet $-s\zeta(s+1, a)$ is exactly $\text{div } \zeta(\cdot, a) - 1$, with multiplicities, in the whole plane; that of the m -th jet is $(\text{div } \zeta(\cdot, a) - m) + [0] + [-1] + \dots + [2-m]$, the last $m-1$ zeros simple and distinct from the shifted ones. At $a = 1$ the zeros of the first jet in $-1 < \text{Re } s < 0$ are exactly $\mathcal{Z} - 1$, and those of the m -th jet in $-m < \text{Re } s < 1-m$ are exactly $\mathcal{Z} - m$. (d) ζ has no zero on $\mathcal{Z} - 1$.

Proof. (a) Differentiate termwise on $\text{Re } s > 1$ and extend by joint analyticity (Euler-Maclaurin); $(s)_m$ vanishes simply at $1-m$, where $\zeta(s+m, a)$ has a simple pole, and $\zeta(s, a) - 1/(s-1)$ is entire. (b) $s\zeta(s+1, a) = 1 + s(\zeta(s+1, a) - 1/s)$. (c) Off $s = 0$ the factor $-s$ is a unit; at $1-m$ a simple zero meets a simple pole of residue 1, and at $-j$, $0 \leq j \leq m-2$, the other factor is $\zeta(m-j, a) > 0$; at $a = 1$, $\text{div } \zeta = \mathcal{Z} + \sum_{k \geq 1} [-2k]$. (d) ζ has no zeros in $-1 < \text{Re } s < 0$. ■

Status. Verified: proved here; the t -derivative at $\rho_k - 1$ is below $4 \cdot 10^{-29}$ for $k = 1, 2, 3$, and check Z1 finds the jet vanishing at $s_0 - 1$ for the zero $s_0 = 1.40778804 + 23.32798775i$ of $\zeta(\cdot, 2)$. Part (a) on $\text{Re } s > 1$ is the programme's shift-flow theorem.

The Hurwitz flow does not translate the zero set; what vanishes on the translate, at every time of the flow, is its first jet. Theorem 4.2 uses the functional equation of ζ and does not extend to other sheets: at $a = 1/2$, $s_1 = 2\pi i/\log 2$ is a zero of $\zeta(s, 1/2) = (2^s - 1)\zeta(s)$, while $\zeta(1 - \bar{s}_1, 1/2) = \zeta(1 + 2\pi i/\log 2) \neq 0$, so the zero set of that sheet is not symmetric under $s \mapsto 1 - \bar{s}$. Part (d) is proved from the functional equation and is not claimed for other sheets.

Theorem 4.2 (the shifted set is the mirror set). $A(\mathcal{Z}) = \mathcal{Z} - 1$ with multiplicities, unconditionally, and $A(\rho) = \rho^{\#} - 1$. Since $A(s) - (s-1) = 1 - 2\text{Re } s$, $A(\rho) = \rho - 1$ exactly when $\text{Re } \rho = 1/2$, so $\text{RH} \iff A(\rho) = \rho - 1$ for every nontrivial zero.

Proof. $-\bar{\rho} = (1 - \bar{\rho}) - 1$ with $1 - \bar{\rho} \in \mathcal{Z}$ of the same multiplicity. ■ *Status.* Verified: proved here.

So the author's two operations produce the same line: as sets the mirror and the shift agree unconditionally, as maps on the zeros they agree exactly under RH, and ζ does not vanish at any point of $\mathcal{Z} - 1$.

Proposition 4.3 (bounded dilation-equivariant maps between weights). Let $H_\sigma = L^2((0, \infty), u^{2\sigma-1} du)$ and $U_a f(u) = f(au)$. If $\sigma \neq \sigma'$, every bounded $K : H_\sigma \rightarrow H_{\sigma'}$ with $KU_{a_0} = U_{a_0}K$ for one $a_0 \neq 1$ is zero. The inversion $f \mapsto f(1/u)$ maps H_σ isometrically onto $H_{-\sigma}$, acting on Mellin transforms by $s \mapsto -s$.

Proof. $U_a = a^{-\sigma} V_a$ with V_a unitary, so $\|Kf\| \leq a^{\{\sigma' - \sigma\}} \|K\| \|f\|$ for $a = a_0^k$, $k \in \mathbb{Z}$, and $k \rightarrow \pm\infty$ gives $Kf = 0$. ■ *Status.* Verified: proved here. Boundedness is needed: the identity on $C_c^\infty(0, \infty)$ is a nonzero closable densely defined operator $H_{1/2} \rightarrow H_{-1/2}$ commuting with every U_a .

4.2 The two centres of the explicit formula

With $(g_1 \star g_2)(x) = \int g_1(y)g_2(x/y)dy/y$ and $\hat{g}$ the Mellin transform, use Weil's explicit formula in Bombieri's form [Bo].

Proposition 4.4 (two centres). (a) With $g \star(x) = x^{-1} \text{conj}(g(1/x))$ and $h = g \star g \star$, $\hat{h}(s) = \hat{g}(s) \text{conj}(\hat{g}(1 - \bar{s}))$, both sides of the explicit formula are real, an on-line zero contributes $m_\rho |\hat{g}(\rho)|^2 \geq 0$, and an off-line pair contributes $2m_\rho \text{Re}(\hat{g}(\rho) \text{conj}(\hat{g}(\rho^\#)))$, an indefinite form with eigenvalues $\pm m_\rho$. (b) With $g^\flat(x) = \text{conj}(g(1/x))$, $\hat{h}(s) = \hat{g}(s) \text{conj}(\hat{g}(-\bar{s}))$, the prime term has imaginary part $\sum_n \Lambda(n)(1 - 1/n) \text{Im } h(n)$, and the sum over zeros is not real in general.

Proof. The substitution $y = 1/x$; in (a), $h(n) + h(1/n)/n = 2\text{Re } h(n)$, and in (b), $h(n) + \text{conj}(h(n))/n$. ■ *Status.* Verified: proved here; both sides computed for $g(x) = \exp(-(\log x)^2/2 + (i/2)\log x - i\gamma_1 \log x)$ (centre $1/2$: 2π to $4 \cdot 10^{-15}$; centre 0 : $2\pi e^{i/2}$ to $1.6 \cdot 10^{-12}$).

So a real partner pairing selects the centre $1/2$, and with it the only possible source of negativity is the hyperbolic block of an off-line pair; by Weil's criterion a negative value exists if and only if RH fails. The same substitution shows that the two normalizations of running the clock backwards, $g \mapsto \text{conj}(g(1/u))$ and $g \mapsto \text{conj}(g(1/u))/u$, have Mellin shadows $s \mapsto -\bar{s}$ and $s \mapsto 1 - \bar{s}$.

4.3 The two-line product and the chiral form

Proposition 4.5. $G(-\bar{s}) = \text{conj}(G(s))$, and $G(it) = |\xi(1 + it)|^2/16 > 0$ for real t (since $\xi(it) = \text{conj}(\xi(1 + it))$ and $\zeta \neq 0$ on $\text{Re } s = 1$). *Status.* Verified: proved here; checked to 10^{-30} .

The maps id , A , P ($\#$ on $\mathcal{Z}$ and $s \mapsto -1 - \bar{s}$ on $\mathcal{Z} - 1$) and $T = AP$ form a Klein four-group preserving $\mathcal{Z} \sqcup (\mathcal{Z} - 1)$. An on-line zero has orbit $\{\rho, \rho - 1\}$, an off-line zero the four points $\{\rho, \rho^\#, \rho^\# - 1, \rho - 1\}$ (the author's doublet); the loop across by A and back by T^{-1} has holonomy $\#$, trivial exactly at on-line zeros; and the zeros of G in $-1/2 < \text{Re } s < 1/2$ (the bulk) are the zeros with $\text{Re } \rho < 1/2$ and their mirrors, so the bulk is empty if and only if RH holds (verified; elementary). For f on $\mathcal{Z} \sqcup (\mathcal{Z} - 1)$ put $Q_A(f) = \sum_w m_w \text{conj}(f(Aw))$, the chiral cross form; Q_A and Ω are real.

Proposition 4.6 (the chiral form and Weil's form). Let f be finitely supported, or the restriction of an element of $\mathcal{B}$, and let $u(\rho) = (f(\rho) + f(\rho - 1))/2$ and $v(\rho) = (f(\rho) - f(\rho - 1))/2$ be its T-even and T-odd parts. Then $Q_A(f) = 2\Omega(u) - 2\Omega(v)$, and:

- (a) A has no fixed point, so Q_A is an orthogonal sum of hyperbolic planes, of signature (n, n) on data supported on n pairs $\{w, Aw\}$, whether or not RH holds;
- (b) $\Omega(u) = \sum_{\text{on-line}} m_\rho |u(\rho)|^2 + \sum_{\text{off-line pairs}} 2m_\rho \text{Re}(u(\rho) \text{conj}(u(\rho^\#)))$, so $\Omega \geq 0$ on all finitely supported u exactly when every zero is on the line;
- (c) the T-even and T-odd data V_+ , V_- are Q_A -orthogonal, with $Q_A = 2\Omega$ on V_+ and -2Ω on V_- ; on T-even data supported on a finite $\#$ -closed set with n_{on} points on the line and n_{pairs} off-line pairs the signature is $(n_{\text{on}} + n_{\text{pairs}}, n_{\text{pairs}})$; under RH, V_+ is maximal positive and V_- maximal negative, and if RH fails V_+ contains a negative direction.

Proof. $A\rho = \rho^\# - 1$ and $A(\rho - 1) = \rho^\#$, so $Q_A(f) = \sum_\rho m_\rho [f(\rho) \text{conj}(f(\rho^\# - 1)) + f(\rho - 1) \text{conj}(f(\rho^\#))]$; substituting $f = u + v$ on $\mathcal{Z}$ and $f = u - v$ on $\mathcal{Z} - 1$, the cross terms cancel in pairs. (a) $Aw = w$ means

Re $w = 0$, where $G \neq 0$. (b) $\#$ fixes on-line zeros and pairs the others; take $u = 1$ at ρ and -1 at $\rho^\#$. (c) For T-even u and T-odd v , $Q_A(u, v) = \sum_{\rho} m_{\rho} u(\rho) [-\text{conj}(v(\rho^\#)) + \text{conj}(v(\rho^\#))] = 0$; on-line points give positive squares and off-line pairs hyperbolic planes; under RH, $\Omega(u) = \sum m|u|^2$, and a subspace strictly containing V_+ contains a nonzero $x \in V_-$. ■ *Status*. Verified: proved here; checks B1–B5.

Corollary 4.7 (an RH criterion in two-line terms). For $k \in \mathcal{B}$ put $f_k(s) = F_*(s+1)F_*(s-1)k(s) + F_*(s+2)F_*(s)k(s+1)$. Then f_k is T-even on the zero set, with $u(\rho) = F_*(\rho+1)F_*(\rho-1)k(\rho)$, and $\text{RH} \Leftrightarrow Q_A(f_k) \geq 0$ for every $k \in \mathcal{B}$.

Proof. $F_*(\rho) = 0$ and $F_*(\rho \pm 1) \neq 0$; the programme's sections realize any finitely supported values of k on $\mathcal{Z}$; apply Proposition 4.6(b). ■ *Status*. Verified: proved here; equivalent to RH by construction, since its test data localize at the zeros.

Theorem 4.2 and Proposition 4.6 hold for every primitive Dirichlet L-function, with its zeros Z_χ and GRH for χ , since $\Lambda(1 - \bar{s}, \chi) = W(\chi)\text{conj}(\Lambda(s, \chi))$ makes $\#$ permute Z_χ (the standard inputs for $L(s, \chi)$ are not verified here); Corollary 4.7 is specific to ζ . $Q_A(f)$ is Weil's explicit-formula functional of G applied to $f \wedge A$, $f \wedge A(s) = \text{conj}(f(-\bar{s}))$, with prime coefficients $\Lambda(n)(1 + n^{-1})$ in $-G'/G$.

4.4 Counting in the two-line system

Proposition 4.8 (parity). For $T > 0$ not an ordinate, $(-1)^{\#\{N(T)\}} = (-1)^{\#\{N_0(T)\}} = -\text{sgn } Z(T)$, with no simplicity assumption. The two-line count $n_G(T) = 2N_0(T) + 4P(T)$ vanishes mod 2, and $i^{\#\{n_G(T)\}} = (-1)^{\#\{N_0(T)\}}$; both are functions of $N_0(T)$ alone, and mod 8, n_G gives $P(T) \bmod 2$ only once $N_0(T) \bmod 4$ is known ($(N_0, P) = (0, 1)$ and $(2, 0)$ both give $n_G = 4$).

Proof. $\xi(1/2 + it) = -(1/2)(1/4 + t^2)^{\pi/4} |\Gamma(1/4 + it/2)| Z(t)$; $\xi(1/2) > 0$, and $\xi(1/2 + it)$ changes sign exactly at on-line zeros of odd order, and off-line zeros come in pairs of equal height. ■ *Status*. Verified: proved here; checked at eight heights up to 1000.

Proposition 4.9 (the holonomy index is Turing's). With $P(T)$ the number of zeros with $0 < \gamma \leq T$ and $\text{Re } \rho < 1/2$,

$$I(T) := (N(T) - V(T))/2 = P(T) + \sum_{\{\text{Re } \rho = 1/2, 0 < \gamma \leq T\}} \lfloor m_{\rho}/2 \rfloor.$$

So $I(T) \geq 0$, with equality exactly when every zero up to height T is on the line and simple; a positive value arises from off-line pairs and from multiple on-line zeros alike.

Proof. Off-line pairs contribute $2P(T)$ to $N(T)$, and Z changes sign exactly at on-line zeros of odd order, so $V(T) = \sum_{\{\text{on-line}\}} (m_{\rho} - 2\lfloor m_{\rho}/2 \rfloor)$. ■ *Status*. Verified: proved here; checks D1–D4 and R5 ($N = V = 29$ at $T = 100$, $N = V = 108$ at $T = 250$).

$I(T)$ is computed from boundary data ($N(T) = \theta(T)/\pi + 1 + S(T)$ by the argument principle, $V(T)$ from the line), and $I(T) = 0$ is exactly what verifications of RH by Turing's method [Tu] establish up to a height; RH is verified up to $3 \cdot 10^{12}$ [PT]. $I(T) = 0$ for all T is RH together with the simplicity of all zeros.

Proposition 4.10 (the second line and prime races). For primitive $\chi \bmod q$, the prime coefficients of $-G_\chi'/G_\chi$, $G_\chi(s) = \Lambda(s, \chi)\Lambda(s+1, \chi)$, are $\Lambda(n)\chi(n)(1 + n^{-1})$. For every modulus q and classes $a \not\equiv b$ prime to q (and races among r classes), the two-line races differ from the one-line races by convergent series, which vanish after Rubinstein–Sarnak's normalization; so they have the same limiting logarithmic distributions (under GRH) and logarithmic densities (under GRH and linear independence) [RS, Dev], for example $\delta(4; 3, 1) \approx 0.9959$.

Proof. In $\sum_{\{p \leq x\}} (1_{\{p \equiv a\}} - 1_{\{p \equiv b\}}) \log p/p = \varphi(q)^{-1} \sum_{\{\chi \neq \chi_0\}} (\bar{\chi}(a) - \bar{\chi}(b)) \sum_{\{p \leq x\}} \chi(p) \log p/p$ the principal character cancels, and each inner series converges [Da, Ch. 20]. ■ *Status*. Verified: proved here; checks C1–C3 and for $q \in \{3, 5, 7, 8, 12\}$, up to 10^7 .

4.5 The separator

Lemma 4.11 (the separator with matched rates). Let $G \in \mathcal{B}$ have zero divisor $D_+ \sqcup D_-$ with $D_+ \subset \{\text{Re } s > 0\}$, $D_- \subset \{\text{Re } s < 0\}$. Suppose that, with $r(t) = \varepsilon(1 + t^2)^{-N}$, (a) $|G(x + it)| \leq C_A(1 + |t|)^{k_A} e^{-\alpha|t|}$ on every strip $|x| \leq A$, and (b) G has no zeros in the corridor $\mathcal{K} = \{|x| \leq r(t)\}$, where

$|G| \geq c(1 + |t|)^{-K} e^{\{-\alpha|t|\}}$. Then there is an entire E of polynomial growth on vertical strips with $E - 1$ vanishing to full order on D_+ and E on D_- , so $\mathcal{B}/(I_{D_+} \cap I_{D_-}) \cong \mathcal{B}/I_{D_+} \oplus \mathcal{B}/I_{D_-}$ topologically, via $([f], [g]) \mapsto [Ef + (1 - E)g]$, commuting with s and every $a^\pm$.

Outline. With a cutoff $\chi(x + it) = \eta(x/r(t))$ and $h = \partial\chi/G$, solve $\partial u = h$ by $u = \pi^{-1} \{ -1 \} f e^{\{(s-w)^2\}(s-w)^{-1} h(w) dA(w)}$; then $|u| \leq C_A e^{\{\alpha|t|\}(1 + |t|)^{2N+K}}$, and $E = \chi - Gu$ works because the two exponential rates cancel exactly. *Status.* Verified: proved in the programme for ζ at GSL4-GSL5 and checked in the audit, which stated it with general constants; standard ∂ technique. For ζ , $G = F_-(s)F_-(s + 1)$ and $\alpha = \pi/2$: its only exponential factor, $1/\cos(\pi s/2)$, has the same size on every vertical line.

Proposition 4.12 (the separator for every primitive Dirichlet L-function). Let χ be primitive mod $q \geq 3$ with $\chi(-1) = (-1)^a$, $\Lambda_\chi(s) = (q/\pi)^{\{(s+a)/2\}} \Gamma((s+a)/2) L(s, \chi)$, and $I_\chi, I_{\{\chi, -\}}$ the ideals of full-order vanishing on Z_χ and $Z_\chi - 1$. Then $\mathcal{B}/(I_\chi \cap I_{\{\chi, -\}}) \cong \mathcal{B}/I_\chi \oplus \mathcal{B}/I_{\{\chi, -\}}$ equivariantly.

Proof. We check Lemma 4.11 for $G_\chi(s) = \Lambda_\chi(s)\Lambda_\chi(s + 1)$ with $\alpha = \pi/2$, from the functional equation $\Lambda_\chi(s) = W(\chi)\Lambda_{\bar{\chi}}(1 - s)$, $L(1 + it, \chi) \neq 0$ and Stirling's formula [Da]. Partial summation against periodic antiderivatives gives polynomial growth, so $\Lambda_\chi \in \mathcal{B}$, and $Z_\chi \subset \{0 < \operatorname{Re} s < 1\}$. By the functional equation, $G_\chi(s) = W(\chi)(q/\pi)^{\{1+a\}} \Gamma((1+a-s)/2) \Gamma((1+a+s)/2) L(1-s, \bar{\chi}) L(1+s, \chi)$, whose Gamma product, $\pi/\cos(\pi s/2)$ or $(\pi s/2)/\sin(\pi s/2)$, has modulus at most $C_A(1 + |t|)e^{\{-\pi|t|/2\}}$ on strips and at least $ce^{\{-\pi|t|/2\}}$ in the corridor. The inequality $\zeta(\sigma)^3 |L(\sigma + it, \chi)|^4 |L(\sigma + 2it, \chi^2)|^2 \geq 1$ for $\sigma > 1$ gives $|L(1 + \delta + it, \chi)| \geq c\delta^{\{3/4\}}(1 + |t|)^{\{-1/2\}}$, hence $|L(1 + x + it, \chi)| \geq c(1 + |t|)^{\{-5\}}$ for $|x| \leq c(1 + |t|)^{\{-6\}}$; and $G_\chi(it) \neq 0$ near $t = 0$. So the lemma applies with $r(t) = \varepsilon(1 + t^2)^{\{-3\}}$. ■

Status. Verified: proved here; the matched rates checked on $\operatorname{Re} s = 0$ for two characters. Novelty: not searched.

The inputs of this proof are the functional equation, polynomial growth, Stirling's formula and zero-freeness on $\operatorname{Re} s \geq 1$, which is where the Euler product is used; its conclusion holds for every configuration of zeros in the open strip. For the Davenport-Heilbronn function, which has no Euler product, the hypotheses of Lemma 4.11 fail on every vertical line, since it has zeros with $\operatorname{Re} s > 1$ [SW] and hence with $\operatorname{Re} s < 0$; whether its splitting fails is open (Question 4). The same argument gives the splitting for every Dedekind zeta function, with $\alpha = \pi/2$ (checked for four fields; the standard inputs are not verified here). The derived separation $\operatorname{RHom}(Q, Q_+) \simeq 0 \simeq \operatorname{RHom}(Q_+, Q)$, $Q_+ = \mathcal{B}/I_+$ with jets at $\zeta + 1$, is the vanishing of Ext for a central element acting as 0 and as 1, and $\mathcal{B}$ itself does not split along the two lines, Q_+ being torsion and $\mathcal{B}$ torsion-free (GMS; verified).

4.6 The defect ratio

For $r > 1$ the programme's defect has eigenvalue $d_r(\rho) = r^{\{\bar{\rho}\}} - r^{\{1-\rho\}} = \operatorname{conj}(r^{\{\rho\}} - r^{\{\rho^\# \}})$ at a zero: a zero minus its reflected partner.

Lemma 4.13. For $r \neq t > 1$, $0 < \beta < 1$, $\beta \neq 1/2$, $x = \beta - 1/2$: $|d_r(\rho)/d_t(\rho)| = \sqrt{(r/t) \cdot \sinh(x \log r) / \sinh(x \log t)}$, strictly increasing in $|x|$ on $(0, 1/2)$ if $r > t$ and decreasing if $r < t$, with values filling the open interval between $\sqrt{(r/t) \cdot \log r / \log t}$ and $(r - 1)/(t - 1)$.

Proof. $r^\beta - r^{\{1-\beta\}} = 2\sqrt{r} \sinh(x \log r)$, and $c \mapsto c \coth(cx)$ is increasing for $x > 0$. ■ *Status.* Verified: proved here; it sharpens the programme's AST6.2; a statement about admissible positions for any β . The programme's further realization of the two lines, paired with their shifts, as the two sheets of a connected double cover of the equator of $\mathbb{CP}^1$ with deck transformation the central -1 of $\operatorname{SU}(2)$ was found in the audit and is not verified here.

5. Weights: Deligne's mechanism against the programme

A large part of the programme, including a full reader of Deligne's *La conjecture de Weil. II* [De], asks which steps of Deligne's proof have analogues for ζ . The weight of α relative to a prime p is $w(\alpha) = 2 \log|\alpha|/\log p$; a zero ρ enters the local data as the eigenvalue p^ρ , of weight $2\operatorname{Re} \rho$. For a conjugation-stable finite set S of zeros the tensor-power trace $(\sum_{\rho \in S} m_\rho n^\rho)^k$ (TWC7) is real, so its even powers are nonnegative: this is Deligne's positivity step (Weil II 1.5; DP4 of the programme's Deligne reader).

5.1 Positivity of even tensor powers

Proposition 5.1 (the exponent data return the largest real part). Let S be a finite set of complex numbers with positive integer weights m_ρ , $\beta_{\max} = \max \operatorname{Re} \rho$, $k \geq 1$, and $D_S^{\wedge\{k\}}(s) = \sum_{n \geq 1} (\sum_{\rho} m_\rho n^\rho)^{\wedge k} n^{-s}$. Then $D_S^{\wedge\{k\}} = \sum_w c_w \zeta(s - w)$ over the distinct sums w of k elements of S , with every $c_w > 0$, and its abscissa of convergence and of absolute convergence is exactly $1 + k\beta_{\max}$. For even k and conjugation-stable S ($m_{\bar{\rho}} = m_\rho$) the coefficients are nonnegative and $1 + k\beta_{\max}$ is a pole with positive residue. So a bound $1 + k/2 + C$ on the abscissa, uniform in k , holds if and only if $\beta_{\max} \leq 1/2$.

Proof. Expanding the power, each $\zeta(s - w)$ converges absolutely on $\operatorname{Re} s > 1 + \operatorname{Re} w$, and convergence on $\operatorname{Re} s > \sigma_1$ with $\sigma_1 < 1 + k\beta_{\max}$ would make $\sum c_w \zeta(s - w)$ holomorphic there, although it has a pole with residue $c_w > 0$ at each $1 + w$ (the w are distinct), among them $1 + kp$ with $\operatorname{Re} \rho = \beta_{\max}$; for even k , $k\beta_{\max}$ is a sum of $k/2$ copies each of ρ and $\bar{\rho}$. ■ *Status.* Verified: proved here; found in the audit (*new*), in every degree and for every finite S (check Z5).

So the exponent data return $\max \operatorname{Re} \rho$ in every degree, positive or not; positivity of even powers (Landau's theorem) places a pole at the real point, and a bound $1 + k/2 + C$, being equivalent to $\max \operatorname{Re} \rho \leq 1/2$, has to come from another source. For odd k the real point need not be a pole ($S = \{0.7 \pm 10i, 0.3 \pm 10i\}$, $k = 3$: the rightmost poles are $3.1 \pm 10i$ and $3.1 \pm 30i$). Deligne's next step (Weil II 1.5; DP5) is exactly an independent bound of the form $k + 1 + C$, from the H^2_c of a fixed curve.

Proposition 5.2 (the average weight of an off-line quartet). Let ρ be a zero of multiplicity m and V_ρ the span of the jet blocks of the distinct points among $\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}$. Off the line V_ρ has rank $4m$ and $\det(W_\rho | V_\rho) = p^{\wedge\{2m\}}$; on the line ($1 - \rho = \bar{\rho}$) it has rank $2m$ and determinant $p^{\wedge m}$. In both cases the weight of the determinant divided by the rank is 1, whatever $\operatorname{Re} \rho$ is, while the constituent weights $2\operatorname{Re} \rho$ and $2(1 - \operatorname{Re} \rho)$ differ from 1 exactly when ρ is off the line. *Proof.* $p^{\wedge\{(\rho + \bar{\rho} + (1 - \rho) + (1 - \bar{\rho}))m\}} = p^{\wedge\{2m\}}$ and $p^{\wedge\{(\rho + \bar{\rho})m\}} = p^{\wedge m}$. ■ *Status.* Verified: proved here; proved in the programme at DW8.12 (the on-line case corrected in the audit; check Z6).

Deligne's Théorème 1.5.1 turns a pure determinant weight of each constituent into purity. For the quartet the four characters are separate constituents, with determinant weights $2\operatorname{Re} \rho$ and $2(1 - \operatorname{Re} \rho)$, and 1.5.1 applies with a conclusion that holds for every value of $\operatorname{Re} \rho$; if they formed one ι -real constituent of rank 4, its determinant weight would be 1 and 1.5.1 would force $\operatorname{Re} \rho = 1/2$. The input this route needs is one that forces the constituents onto the average weight 1, equivalently a pole bound at $k + 1 + C$ uniform in k : (a) a square with a Künneth-type map $H^1 \otimes H^1 \rightarrow H^2$ and an auxiliary object with discrete weights, or (b) a cohomological source for the uniform bound, which on the finite data is equivalent to $\max \operatorname{Re} \rho \leq 1/2$ and so has to be independent, as the H^2_c of a fixed curve is in Weil II §3.2. TWC9 and TWC11 state that no construction of the programme has yet been assigned the fixed-curve denominator and that its computations do not rule one out.

5.2 An algebraic purity criterion

Let V be finite-dimensional, N nilpotent and F invertible with $FN F^{\wedge\{-1\}} = Q^{\wedge\{-1\}}N$, $|Q| \neq 1$, $w(\alpha) = 2 \log|\alpha|/\log|Q|$, M the monodromy filtration of N , and $F^{\wedge v} = (F^{\wedge\{-1\}})^{\wedge t}$, $N^{\wedge v} = -N^{\wedge t}$ on $V^{\wedge v}$.

Proposition 5.3 (the core of Weil II 1.8.4). These are equivalent: (a) for every i , all eigenvalues of F on $\operatorname{Gr}^{\wedge M}_i V$ have weight $\beta + i$; (b) all eigenvalues of $F \otimes F$ on $\ker(N \otimes 1 + 1 \otimes N)$ have weight $\leq 2\beta$, and all eigenvalues of $F^{\wedge v} \otimes F^{\wedge v}$ on $\ker(N^{\wedge v} \otimes 1 + 1 \otimes N^{\wedge v})$ have weight $\leq -2\beta$. The bounds on $\ker N$ for V and $V^{\wedge v}$ alone do not suffice.

Proof. Inputs: Weil II (1.6.6), (1.6.9), (1.6.14), with the twist of (1.6.14.3) read as $-(i + j)/2$ (§5.4). (a) $\Rightarrow$ (b): the monodromy filtration of $N \otimes 1 + 1 \otimes N$ is $\sum_{a+b=k} M_a \otimes M_b$, so its Gr_k has weight $2\beta + k$, and the kernel lies in M_0 ; dually $\operatorname{Gr}_i(V^{\wedge v}) \cong (\operatorname{Gr}_{\{-i\}} V)^{\wedge v}$. (b) $\Rightarrow$ (a): for an eigenvalue α on the primitive part $P_{\{-j\}}$ with eigenvector $N^{\wedge j}y$, $y \in \operatorname{Gr}_j$ (so $Fy = \alpha Q^{\wedge j}y$), the vector $\sum_{s \geq 0} (-1)^s N^{\wedge s}y \otimes N^{\wedge\{j-s\}}y \in \operatorname{Gr}_0(V \otimes V)$ is nonzero, is killed by $N \otimes 1 + 1 \otimes N$, and has eigenvalue $\alpha^2 Q^{\wedge j}$, so $2w(\alpha) + 2j \leq 2\beta$; dually $w(\alpha) \geq \beta - j$; so $P_{\{-j\}}$ has weight $\beta - j$, and $\operatorname{Gr}_i \cong \bigoplus_j P_{\{-j\}}(-(i + j)/2)$ has weight $\beta + i$. For the last claim, $V = \langle v \rangle \oplus \langle v_1, v_0 \rangle$ with $Nv_1 = v_0$, $Nv_0 = Nv = 0$, $Fv = av$, $Fv_1 = bv_1$, $Fv_0 = (b/Q)v_0$, $w(a) = \beta$ and $w(b) = c + 1$: the kernel bounds hold for $\beta - 1 \leq c \leq \beta + 1$, (a) only for $c = \beta$, and $v_1 \otimes v_0 - v_0 \otimes v_1$ has weight $2c$. ■

Status. Verified: proved here and checked on 142 random models. (b) $\Rightarrow$ (a) is the algebra of Deligne’s proof (Weil II p. 176); the converse and the example were added in the audit; the equivalence is not claimed new. In Weil II, (b) comes from geometry (Lemme 1.8.1). An operator commuting with N , such as the programme’s counting operators W_p and C_b , has the same eigenvalues on Gr_i and Gr_{-i} , so its weights are of the form $\beta + i$ only if $N = 0$; the ramification operator P_b satisfies $NP_b = bP_bN$ and has no eigenvector on its space (verified; found in the audit).

5.3 Three lemmas from Weil II §2

Lemma 5.4 (concentration, with the outside term). Let K be a compact group, ϑ an irreducible character of degree d_ϑ , $V \subset K$ with $\text{Re } \chi_\vartheta \geq (1 - \varepsilon_1)d_\vartheta$ on V , and $\rho_0 \neq 0$ an integer combination of characters with $\int_{K \setminus V} |\rho_0|^2 \leq \varepsilon_2 \int |\rho_0|^2$; write $\rho_0 = \rho^+ - \rho^-$ with genuine $\rho^\pm$ without common constituents and $\rho = \rho^+ + \rho^-$. Then $[\rho\bar{\rho} : \vartheta] \geq ((1 - \varepsilon_1)(1 - \varepsilon_2) - \varepsilon_2)d_\vartheta[\rho\bar{\rho} : 1]$, with $[\rho\bar{\rho} : 1] = \int |\rho_0|^2$, sharp in ε_2 . The printed choice $\varepsilon_1 + \varepsilon_2 < \varepsilon$ in the proof of Weil II 2.1.7 does not give Deligne’s inequality; $\varepsilon_1 + 2\varepsilon_2 < \varepsilon$ does.

Proof. $\rho\bar{\rho} = \rho_0\bar{\rho}_0 + 2(\rho^+\bar{\rho}^- + \rho^-\bar{\rho}^+)$, and $[\rho_0\bar{\rho}_0 : \vartheta] = \int |\rho_0|^2 \text{Re } \chi_\vartheta$, whose part over V is at least $(1 - \varepsilon_1)(1 - \varepsilon_2)d_\vartheta \int |\rho_0|^2$ and whose part off V is at least $-\varepsilon_2 d_\vartheta \int |\rho_0|^2$. On $\mathbb{Z}/2$ with $\vartheta = \text{sgn}$, $V = \{e\}$ and $\rho_0 = 23 + 9\text{sgn}$: $\int |\rho_0|^2 = 610$, off- V fraction $98/610 < 0.18$, and with $\varepsilon_1 = 0.01$, $\varepsilon_2 = 0.18$, $\varepsilon = 0.2$ one gets $[\rho\bar{\rho} : \text{sgn}] = 414 < 0.8 \cdot 610 = 488$. ■ *Status.* Verified: proved here; found in the audit (*new*), the repair being the programme’s Deligne reader’s (DB0, DB5); checks B1–B4.

Lemma 5.5 (integer positivity; Weil II 2.1.5–2.1.6). If $\nu : \hat{K} \rightarrow \mathbb{R}$ has $\nu(1) = 1$, $\nu(\vartheta) \leq 0$ for $\vartheta \neq 1$ and $\nu(\rho\bar{\rho}) \geq 0$ for every genuine ρ , then $\sum_{\vartheta \neq 1} d_\vartheta(-\nu(\vartheta)) \leq 1$ (Lemma 5.4 with Peter-Weyl, letting $e \rightarrow 0$). For integer-valued ν with $\nu(\vartheta) = \nu(\vartheta^2)$, at most one $\vartheta \neq 1$ has $\nu(\vartheta) < 0$, a quadratic character with $\nu(\vartheta) = -1$, and on $\mathbb{Z}/2$, $\nu(\text{sgn}) = -1$ satisfies every hypothesis; on $\mathbb{Z}/3$ with $\nu(\vartheta) = \nu(\vartheta^2) = -x$, positivity holds if and only if $x \leq 1/2$, since $\nu(\rho\bar{\rho}) = a^2 + b^2 + c^2 - 2x(ab + bc + ca)$. *Status.* Verified: proved here; found in the audit (*new*). In Weil II §2.2 the exception is removed with a double cover; for ζ ($G = \mathbb{R}$, $K = \{0\}$, Deligne’s example 2.1.9) there is no nontrivial quadratic character, and the method gives exactly $\zeta(1 + it) \neq 0$.

Lemma 5.6 (nonnegative power sums). If $a_j \in \mathbb{C}^\times$ and real δ_n with $\limsup \max(\delta_n, 0)^{1/n} \leq D$ satisfy $\delta_n - \sum_j a_j^n \geq 0$ for all n , then $\max |a_j| \leq D$, since Dirichlet’s simultaneous approximation makes the largest terms nearly real and positive for infinitely many n ; for a curve over $\mathbb{F}_q$, $\#X(\mathbb{F}_{q^n}) \geq 0$ alone gives $|\alpha_j| \leq q$. Hence (**Corollary 5.7**) the zeta function of a smooth projective geometrically connected curve over $\mathbb{F}_q$ has a simple pole at $s = 1$ by the point counts alone, since an eigenvalue q on H^1 would make the counts bounded. *Status.* Verified: proved here; Lemma 5.6 found in the audit (*new*); Corollary 5.7 proved in the programme at DBC7.

5.4 Misprints in the printed Weil II

The programme’s Deligne reader flagged these displays as defects of its transcription; the audit found them in print (the numdam scan, pp. 150–203). None affects a conclusion of Weil II.

Display	Page	Printed	Meant
(1.3.10)(iv)	160	centre onto a finite subgroup of $\mathbb{Z}$	a subgroup of finite index
(1.6.7)	166	exception $i \neq d$	$i \neq -d$
(1.6.13)	169	$k - 2i - 2 \geq k \geq 2j - k$	$k - 2i - 2 \geq -k \geq 2j - k$
(1.6.14.3)	170	$P_{-j}((i + j)/2)$	$P_{-j}(-(i + j)/2)$
(1.7.5), filtration	171	$M'_i = \prod_{j < i} V'_j$	$j \leq i$
(1.7.5), coefficient	171	$\mu = \lambda/(1 - q^n)$	$\mu = \lambda/(1 - q^{-n})$
(2.1.1)	187	period $2\pi i \log q$	$2\pi i / \log q$
proof of 2.1.7	191	$\varepsilon_1 + \varepsilon_2 < \varepsilon$	$\varepsilon_1 + 2\varepsilon_2 < \varepsilon$
2.2.8(i)	195	weight $2\mathcal{R}(\tau)$	$-2\mathcal{R}(\tau)$
(3.1.3.3)	199	Kummer class of uv^{-1} a generator	twice a branch class
proof of 3.2.1	200	$F^v(-1)$	$F^v(1)$
(3.2.7.1)	202	H^1	H^1_c
proof of 3.2.12	203	$w \leq \beta + i$ in a weight-0 proof	$w \leq i$

Status. Verified against the printed pages; found first by the programme's Deligne reader. A quick errata search for the displays of §1 found none published; no novelty is claimed.

6. Further results

Proposition 6.1 (the two poles differ in kind). For even Schwartz h with $\mathcal{M}_{\text{Sh}}(s) = \int_0^\infty h(v)v^{s-1}dv$, $2\zeta(s)\mathcal{M}_{\text{Sh}}(s) = E_h(s) + h(0)/(s-1) - h(0)/s$ with E_h entire (Poisson summation): the pole at 0 is the lattice's $n = 0$ term, with residue $-h(0)$ for every lattice $c\mathbb{Z}$, and the pole at 1 is the dual lattice's, with residue $(\int h)/c$, so one counts a point and the other measures a density. If $h(v) = kv^\alpha + O(v^{\alpha+1})$ at 0, $\mathcal{M}_{\text{Sh}}$ has a pole at $-\alpha$ with residue κ , and the trivial zeros cancel the poles of the smooth even model at $-2, -4, \dots$; the conditions $h(0) = 0 = \int h$ remove exactly the two poles, the pole at 1 carrying the average and each zero an oscillation. *Status.* Verified: proved here; found in the audit; checked to 10^{-31} ; classical in content.

In the programme the object at the point where the nonzero numbers stop is τ . By the local statement of Proposition 6.1 (a pole at $-\alpha$ read from the behaviour kv^α of h at 0), a pole at τ would be computed from a local model of the test functions at τ , which the programme has not specified (Question 7). Reading the residue at 0 as presence without an exchanged label ($0 = -0$ is the only non-free orbit of ± 1) is a proposed typed map, not an identification with τ , since the integer 0 has even parity. The author places τ at the pivot of the swing between 0 and 1; on the zeta side that is the reflection $s \mapsto 1 - \bar{s}$ about $1/2$, where RH reads $\rho^\# = \rho$, and centring at 0 differs from it by exactly one shift, $s^\# = T_1(A(s))$.

Lemma 6.2 (the winding at the zeros). For every nontrivial zero $\rho = \sigma + i\gamma$, $\lambda_\rho = e^{\{-2\pi i\rho\}}$ has $|\lambda_\rho| = e^{\{2\pi\gamma\}}$, is a negative real number if and only if $\sigma = 1/2$, and satisfies $|1/(\lambda_\rho - 1) + 1_{\{\gamma < 0\}}| \leq 1/(e^{\{2\pi|\gamma|\}} - 1) < 2.7 \cdot 10^{-39}$ (as $|\gamma| \geq 14.1347\dots$). *Status.* Verified: direct computation; found in the audit.

Lemma 6.3 (escape under ramification). Let Z be any set with $0 < \text{Re } \rho < 1$ and $a > 1$ real. Then $\{k \geq 0 : a^k \rho \in Z\}$ is finite, no nonempty $S \subset Z$ has $aS \subset S$, and only points with $\text{Re } \rho < 1/a$ return. If $a \geq 2$ and $Z = 1 - \bar{Z}$, no reflected pair returns together; for $1 < a < 2$ one can ($a = 3/2$, $\rho = 0.55 + i$). Two points collide, $a\rho = a'\rho'$, only if $\rho/\rho' = a'/a > 0$, which is impossible for distinct points of the critical line. *Proof.* $\text{Re}(a^k \rho) = a^k \text{Re } \rho < 1$, and $\text{Re } \rho < 1/a$ and $1 - \text{Re } \rho < 1/a$ are incompatible for $a \geq 2$. ■ *Status.* Verified: proved here; found in the audit. So the programme's iterated- and reflected-return theorems (BRC, GJN, GJNR) hold for every divisor in the open strip with their other hypotheses.

Proposition 6.4 (every failure of RH is detectable). Let T be a recursively axiomatized theory that proves every true Σ_1 sentence and proves that an explicit Turing machine M halts if and only if RH fails (the 744-state machine of Matiyasevich and O'Rear [Aa, Theorem 9], or any machine searching for a counterexample to a Π_1 form of RH [DMR]; ZFC is an example). If RH is false, $T \vdash \neg\text{RH}$; so if RH is independent of T it is true. If T is Σ_1 -sound, then $\text{RH} \iff T \not\vdash \neg\text{RH} \iff \text{Con}(T + \text{RH})$. *Proof.* If RH fails, M halts, a true Σ_1 sentence that T proves; if RH holds and $T \vdash \neg\text{RH}$, then T proves the false Σ_1 sentence that M halts. ■ *Status.* Verified: proved here; derived in the audit from the author's argument; standard logic. So the absence of a detectable failure (the consistency of $T + \text{RH}$) implies RH, and every failure has a finite certificate; for Σ_1 -sound T this absence is equivalent to RH, and a consistent T does not prove $\text{Con}(T + \text{RH})$ (Gödel).

Lemma 6.5 (Keller maps give incompressible polynomial flows). If $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is polynomial with $\det DF \equiv c \neq 0$, V is a C^1 field and $U = DF^\wedge\{-1\}(V \circ F)$, then $\text{div } U = (\text{div } V) \circ F$ (by the Piola identity for $\text{adj}(DF)$), U is polynomial if V is, F maps integral curves of U to those of V , and for a polynomial automorphism U is complete if and only if V is. For the map of Alpöge's counterexample to the Jacobian conjecture ($\det DF = -2$; companion record), $\gamma(r) = (z^\wedge\{-1\}, -3z/2, 13z^2/2)$, $z = \sqrt{1 - 8r}$, is an integral curve of U for $V = (2, 0, 0)$ that escapes every compact set as $r \uparrow 1/8$, with $F(\gamma(r)) = (-1/4 + 2r, 0, 0)$. *Status.* Verified: proved here and checked exactly; found in the audit. The workbench's attribution file records that Alpöge's announcement credits Akhil for the question and Fable for the work.

The Gram sandwich. If $\|(\Phi_J - \Phi)x\| \leq \delta\|\Phi x\|$ for all x with $\delta \leq 1$, then $(1 - \delta)^2\Phi^*\Phi \leq \Phi_J^*\Phi_J \leq (1 + \delta)^2\Phi^*\Phi$, by the triangle inequality; $\delta \leq 1$ is needed for the lower bound (verified). The programme's Schur-complement replacements and the Yang-Mills heat semigroup are instances.

7. Catalogue of the remaining results

The standalone results of the audit's register not stated above; register numbers are given for reference. Each row is verified (proved in the programme at the label given and checked in the audit) unless it says otherwise.

No.	Result
S8, S9	On Connes–Consani's signed winding group $(T, J : [T, J] = 1, J^4 = 1) \cong \mathbb{Z} \times \mathbb{Z}/4$ [CCt], with sign $\varepsilon = J^2$ a
S10–S11	The weight-one sign sector over the $F_{\{1^2\}}$ -points is $y^2 = x^3 - x$, with J acting as complex multiplicati
S14, S15, S17	For nonreal ρ at most two primes have p^ρ algebraic; two primes determine the values at the zeros, a
S18	Least-order lifting $m + k + 1$ at a zero of multiplicity m ; the obstruction class $(-1)^{jj!/2 \cdot t^{r-1-j}} \chi_{\mathbb{Z}}$
S32	The clocks $\mathbb{Z}/1, \dots, \mathbb{Z}/N$ have joint states $\mathbb{Z}/\text{lcm}(1, \dots, N)$, with $\log \text{lcm}(1, \dots, N) = \psi(N)$; for every integ
S36, S70	On the three-point base Y the residue pairing is the trace of a sheaf map on $Y \times Y$, kept by $R\Delta^! = Rq$
S61, S54	The residue extension splits after evaluation at $s_0 \neq 0$ and at $s_0 = 0$ is $n \mapsto n(I - \log n \cdot N)$, with the ten
S16, S58, S62	Deligne's specialization argument with weight bookkeeping; two conditions implied by RH read off th

Part B. Proved negative results

8. Negative results, with their scope

Each result below is a theorem, or where marked a numerical observation, about a specific construction as it is defined, and it is stated with the scope in which it is proved. Where the source identifies the additional input the construction would need, that input is stated; whether the programme's constructions can supply those inputs is open. Derivations are in Part A at the places cited, or in the programme at the labels given, checked in the audit. Three kinds occur: (i) statements that are false, with a proof or counterexample; (ii) constructions whose conclusions hold for every divisor in the strip with the stated properties (or for curves over finite fields, or for functions without an Euler product that have off-line zeros), so that deciding the location of the zeros through them requires an input that distinguishes ζ , such as the Euler product; (iii) constructions equivalent to RH by construction, which isolate the off-line part of the zero set in their definition, so that proving RH through them requires an independent argument that this part vanishes. The inputs identified are a pole bound uniform in the tensor degree, or a discreteness input with a square (the first is equivalent on the programme's finite data to $\max \text{Re } \rho \leq 1/2$, so it would have to come from an independent source); a property that distinguishes ζ from the Davenport–Heilbronn function, such as the Euler product; and polynomial bounds on the principal parts of $1/\zeta$ at the zeros. Bracketed numbers are those of the audit register.

8.1 Statements that fail

- **N-Vel [1].** Under (H), $\sum_{\gamma \leq T} \text{Re } c(\rho) = T/(4\pi) + O(T)$ is false, since $\limsup |\text{Re } E(T)|/T = \infty$, and the vertical law fails likewise (Theorem 2.2(b)); only the smoothed laws hold. Whether the observed skew ($\text{Re } E/T$ between -3.15 and $+0.17$ up to $T = 64\,000$; numerical) persists is open.
- **N-Sheet [3].** For $a \in (0, 1) \setminus \{1/2\}$, $\zeta(s, a)$ has infinitely many zeros with $\text{Re } s > 1$ [DH, Cas], and $\zeta(s, 2) = \zeta(s) - 1$ vanishes at $1.40778804065 + 23.3279877546i$. So the properties shared by all sheets hold for sheets with zeros in $\text{Re } s > 1$, and a proof that the zeros of ζ lie on the line must use a property of the Eulerian sheets $a \in \{1/2, 1\}$, such as the Euler product.
- **N-Beurling [12, 13].** Beurling systems can have $N(x) = kx + O(x^{1/2} \exp(c(\log x)^{2/3}))$ and zeros approaching $\sigma = 1$ [Zh, DMV], and $(\text{primes} \setminus \{2\}) \cup \{3', 4\}$ has $N(x) = x + O(\log^2 x)$ (found in the audit), while exact counts determine the system. The programme's question is how the additive counting step constrains the multiplicative system.
- **N-Free [28].** $\{1\} \cup \{2^e : e \geq 2\}$ is not free although its zeta function has all its zeros on $\text{Re } s = 0$ (Example 2.6); for congruence monoids zero-freeness in $\text{Re } s > 1$ does imply freeness (Proposition 2.7).

- **N-Eu [50].** No finite set of primes fixed in advance has Euler factors that determine every finite shifted-zeta product: for $a_j = 2\pi i / \log p_j$, the even and odd sub-sums of $\{\Sigma_{j \in J} a_j : \varepsilon \in \{0, 1\}^k\}$ give products with equal Euler factors at every p_j and different exponent measures, whose difference has Laplace transform $\prod_j (1 - e^{a_j z}) \neq 0$ (found in the audit).
- **N-Mirror [37].** The second line is $A(\zeta) = \zeta - 1$, and mirror equals shift pointwise if and only if RH holds (Theorem 4.2). The properties used have counterparts for curves over finite fields, where RH is Weil's theorem [We48, Ro], so their consequences hold in a setting where every zero is on the line; the 0-centred pairing is not Hermitian (Proposition 4.4(b)); and the two lines coexist equally for the Davenport–Heilbronn function, which has the zero $0.808517182456637 + 85.6993484853776i$ off the line. To decide the location of the zeros, this route needs a property that distinguishes ζ , such as the Euler product.
- **N-Logic [43].** RH cannot be independent of ZFC and false (Proposition 6.4); for Σ_1 -sound theories, non-refutability of RH is equivalent to RH.

8.2 Positive forms and receivers are supported on the critical zeros

- **N-Pos1 [6, 26].** Every continuous positive form compatible with the scaling transfer vanishes on the off-line part $R = Q/N_O$ (CPS4.2; Lemma 3.13); CFP, CPS, TWC10 and SPF9 are related results; in tensor degree $k \geq 2$ reflected pairs survive, with modulus exactly n . An argument through such receivers needs an additional input that removes R .
- **N-Pos2 [24, 25, 30, 33, 35].** The summation image is dense in $L^2(du)$, so the naive L^2 quotient seminorm is zero (ASD10, OPD3); the swept distribution descends if and only if there are no off-line zeros (HSW6); the Hilbert receivers are pure, with kernel exactly the off-line jet classes (WHR8–WHR12); T_p -invariant seminorms vanish on every primary block with $0 < \operatorname{Re} \rho < 1$ (ATG6.5); and $\Theta_{\text{off}} = \Sigma_{\{\operatorname{Re} \rho \neq 1/2\}} m_{\rho F}(\rho)$ is never a nonzero multiple of $b \mapsto b(1)$ (an explicit b_* has $b_*(1) = 2$ and $\Theta_{\text{off}}(b_*) = 0$; VWR10.5–10.9). My assessment: the spectral-radius argument proves purity of an image that retains only critical-line jets by construction, so to reach RH it would need an input showing that the kernel, the space of off-line jet classes, is zero.
- **N-Pos3 [47, 48, 53, 54, 56].** Equivalent by construction to RH, or to RH with simplicity: the objects and detectors of AST ($\mathcal{R}$), GDC (N_O), PTQ (H_{off} , the descent of the Weil form, PTQ8.3) and ECR5.5; the conditions $W \geq 0$, $U_a = D_a^*$ and the density of the trace image (RTT, RZ); and the vanishing or normality conditions of GDE, ABH ($\mathcal{C}_{\{r,t\}}$), HSR ($D_{\{r,t\}}$, Ξ) and HBW, with unitarity of $n^{-1/2} A_n$, weight-1/2 counting moduli, monodromy phases -1 and $D_{\{r,t\}} \subset H^2$. The two numerical detectors are not holomorphic in ρ ; every attempted cancellation leaves the off-line sector intact; the unconditional results among them (NHI5.4, NHI8.4, HSR4.1–HSR10.6, HBW3, HBW8, HBW9) hold whether or not RH does; and on the Hardy boundary receiver the adjoint-domain condition is met only by $x = 0$ [No].
- **N-Pos4 [29].** Under $s \mapsto 1 - \bar{s}$ the odd part of $\log|\zeta|$ is $(1/2)\log|\chi|$, accounted for by the pole and the trivial zeros, so the detector $\mathcal{E}_r(t)$ is carried by the even part, and the functional equation fixes only the odd part (OZD5, OZD7). My assessment: a zero at height γ enters with the factor $e^{\{2t(\sigma^2 - \gamma^2)\}}$, so for fixed (r, t) zeros above modest heights contribute below any practical numerical precision.

8.3 Constructions whose conclusions hold for a class of divisors

- **N-Unif1 [14, 15, 16].** The inference that an off-critical zero would go undetected is false: $\varphi(v)e^{\{-(\rho-1/2)v\}}$ detects any zero. An off-critical quartet would enter the absorption measure as $-2m(e^{\{\beta t\}} + e^{\{(1-\beta)t\}})\cos \gamma t$, and excluding it requires a sign or growth bound beyond completeness (IH2–IH4). Order-preserving clock changes are linear and move the critical line with the zeros (PL2–PL8).
- **N-Unif2 [18, 19, 20, 22].** The least lift of the k -th jet has order $m + k + 1$ at every zero of multiplicity m , on or off the line (MCL6), and the fixed-order class of the restriction row is nonzero at every zero (ORE5.8, ORE6.5); MCL3–MCL7, ORE6.2, ORE6.5, IH6.4, the first two laws of IH6.5 and PL2.1 hold word for word for $Z_{\{1/5\}}$, which has zeros in $\operatorname{Re} s > 1/2$.
- **N-Unif3 [36, 55].** The cover-invariant part of the zero-coordinate slice is exactly the divisor that the chosen residue section inserts (the zeta divisor for ξ , any finite set for a polynomial section, the empty divisor for a constant; RSS3.2–RSS3.3, RSS6), and two sections of one extension have cocycle spans

- I_{ζ} and $\mathcal{B}$ (NJS10); locating the zeros this way requires an input about ξ beyond the choice of section.
- **N-Unif4 [38, 44, 46].** The two-line splitting and the derived separation hold for every primitive Dirichlet L-function and every Dedekind zeta function, for every configuration of zeros in the open strip (§4.5). The statements involving RH restate it through $\mathcal{R} = Q/N_O$, which is determined by the values at off-line zeros, and the programme states that neither a vanishing nor a new pure weight follows (ADM9).
 - **N-Unif5 [58, 59].** BML's winding difference is invertible for every zero set avoiding the integers with $e^{-2\pi i p}$ away from 1 (Lemma 6.2); BML11's H^1 is produced by truncating the logarithmic tower; the return theorems of BRC, GJN and GJNR hold for every divisor in the strip (Lemma 6.3); $NP_b = bP_bN$ is the chain rule and has solutions for every nilpotent.
 - **N-Unif6 [69, 70, 71, 72].** The doubled source has the cohomology of Connes–Consani's base (§7, S70). The gluing and the pullback (CGS4–CGS10, DCP4–DCP12) hold for every zero set, which enters as the spectrum of the dilations on H^1 and through the jets, and finite models with zeros on and off the line agree in all dimensions, ranks, signs and sections; the lift in DCP vanishes by Poisson summation rather than by a separation of weights (DCP12); the Ext-vanishing with the endpoint modules follows from $\xi(0) = \xi(1) = 1/2 \neq 0$ (GEX9); OMS5–OMS7A hold for every divisor with $0 < \text{Re } \rho < 1$ and unbounded heights (OMS8).
 - **N-Prim [73].** Convergence of the primary decomposition is equivalent to polynomial bounds on the principal parts of $1/\zeta$ at the zeros (Theorem 3.9), the input this construction needs; Q is not the product of its primary blocks, and the constant tuple 1, behind the programme's class c_{ζ} (PGD4), is not a jet tuple.

8.4 Weights, tensor powers and the square

- **N-W1 [17, 21, 65].** The programme's receivers separate weights by the strip bounds or by Tate shifts built into them; a geometric source for a dividing line at weight 1 has not yet been constructed (TL8). Positivity of the timed-prime measure gives $\zeta(1 + it) \neq 0$ and the classical zero-free region, and no zero-free strip (Beurling systems with positive prime measure can have zeros approaching $\text{Re } s = 1$); the compact form, the cohomological proof of condition (C), the strict bound and Weil II §3 are the steps that would need ζ analogues.
- **N-W2 [23, 27, 31, 32, 34, 52, 66].** The exponent data return $\max \text{Re } \rho$ in every tensor degree (Proposition 5.1), and the needed input is a pole bound at $k + 1 + C$ uniform in k . In the programme's square and tensor powers the characters of source and target match, the positive forms kill every tuple off the centred set, the tensor Weil form is indefinite on the reflected sector, and the square's obstruction vanishes exactly when the original one does (TWC6, TWC9–TWC11, FST3–FST7); in degree 1 the square reduces to $Q \otimes Q$ (GMC4–GMC6), and the finite-trace exponent is $k \cdot \max \text{Re } \rho$ (ATG). The tensor Euler products of an off-line quartet have their pole at $1 + 2k \cdot \max(\text{Re } \rho, 1 - \text{Re } \rho)$ for every complex ρ , zero of ζ or not (checked at $0.7 + 10i$); detecting zeros through them would require an input that singles out the zeros of ζ .
- **N-W3 [60], N-Tr [39].** The average weight of an off-line quartet is 1 (Proposition 5.2); Weil II 1.5.1 takes a pure determinant weight as its input, and at a point the weight of the determinant is the sum of the constituent weights. The product trace does not force weight 1: reflected-pair classes have character p whether or not either zero is on the line (FEM7–FEM8), and the diagonal return pairs $2\text{Re } \rho$ with $2 - 2\text{Re } \rho$ (DER10); no degree-zero same-site lift exists (FTD9), and $RA^{\wedge!}K \rightarrow \Delta^{\wedge\{-1\}}K$ is zero (DER6).
- **N-W4 [51, 61, 62, 63].** The bounds on $\ker N$ for V and $V^{\wedge v}$ alone do not give Weil II 1.8.4 (Proposition 5.3); counting operators commuting with N have Deligne's local weights only if $N = 0$; P_b has no eigenvector, so Deligne's weight filtration is not defined for it; and tensor Euler products agree for a non-split Jordan block and its split sum, such as the residue extension at $s = 0$ (ETR10, NPE4).
- **N-W5 [67, 68].** The recovered ring $\text{End}(G/G_{\text{tors}}) \cong \mathbb{Z}$ is formed from the free quotient of the winding group, every automorphism acts trivially on it, and the torsion, the chart twist, the involution and the support τ leave it unchanged; all Frobenius weights of the objects built on it from programme data are even. The purity circle $|a|^2 = p$ needs weight 1, which Deligne's theory supplies through H^1 of an elliptic curve over $\mathbb{F}_p$; a weight-one object built from programme data is the input this route would need.
- **N-W6 [64], N-F1 [7–11].** Deligne's quadratic exception satisfies every positivity hypothesis (Lemma

5.5), and for ζ it is excluded because $\mathbb{R}$ has no nontrivial quadratic character. Doubling has no sign-preserving mirror-compatible lift (§7, S8); the compact-completion modulus theorem holds for every compact group (P12); the $\sqrt{n}$ dilation radius cancels in the product formula; the twistor involution fixes the radius only on $|z| = 1$; and the weight-one sign sector is $L(E, s)$ of conductor 32, which is not ζ .

8.5 The two-line system and the transfers

- **N-Two1 [40].** The $\mathbb{Z}_2$ anomaly of one line is the parity of $N_0(T)$; in the two-line system the $\mathbb{Z}_2$ anomaly vanishes and the $\mathbb{Z}_4$ anomaly is the one-line parity; n_G determines P only together with N_0 ; and the holonomy index counts off-line pairs and multiple on-line zeros together (Propositions 4.8, 4.9).
- **N-Two2 [41, 45].** The Davenport–Heilbronn function has the completed functional equation, the reality, a nonnegative two-line product on $\text{Re } s = 0$ and a four-point orbit for its off-line zero; the hypotheses of the separator lemma fail for it on every vertical line, and whether its splitting fails is open.
- **N-Two3 [74].** For products Λ of completed Dirichlet L -functions containing χ and $\bar{\chi}$ equally often, $G_\Lambda(it) = |\Lambda(1 + it)|^2 > 0$ follows from Λ 's functional equation, reality and nonvanishing on $\text{Re } s = 1$, and G_Λ with its explicit formula is a function of Λ ; so a bound on the off-line count obtained through G_Λ is one obtained from Λ , and whether one-line inputs can bound that count better than Turing's method is open. The chiral form is hyperbolic whether or not RH holds, and its T-even part 2Ω is indefinite exactly when an off-line zero exists; the two-line prime races have the one-line distributions (Proposition 4.10).
- **N-X [42, 49, 57].** Joint averages do not survive restriction of the Navier–Stokes torus cover to unchanged finite torsion (13 against 4 at $p = 13$, $a = 4$). The Jacobian $\rightarrow$ Yang–Mills chain gives statements along couplings $g_j < 1/j$ tending to 0: Theorem 14.2 of the Yang–Mills workbench is correct conditional on fixed-box weak-coupling limits that the file sketches, and its energy quotients are, up to $1 \pm 1/(10j)$, the lowest free two-gluon energy of an open box of side $j/50$, so $Q_j \cdot (\text{box side}) \rightarrow 2\sqrt{2}\pi$, the scaling of a massless free field, for every Jacobian tensor; a statement about the continuum mass gap through this chain would need a fixed-coupling or continuum input. The Navier–Stokes profile gives a zero-quotient sequence only as the coupling tends to 0, and prime-only clock seeds are not transfer-stable.

Part C. Bridges to other areas, and the directions they open

9. Bridges, what is established, and my assessment

Much of the programme consists of bridges between the zeta function and other areas. For each, this part gives the typed map (domain, codomain, what crosses), what is established, what it suggests on both sides, and my assessment, a view held at this stage.

9.1 Meyer's quotient and interpolation theory

Map: the jet map $J : Q \rightarrow \prod_p A_p$; the interpolation problem at the zeros crosses. *Established:* J is injective with dense image in A_m , and onto exactly when the principal parts of $1/\zeta$ are polynomially bounded (Corollary 3.7, Theorem 3.9). *Suggests:* for interpolation theory, the zeta zeros as a test case for interpolating varieties [BT], beside Fourier interpolation at the zeros [BRS]; for ζ , the pointwise form of the questions that the Gonek–Hejhal conjectures ask on average for $\Sigma|\zeta'(\rho)|^{-2k}$. My assessment: this is the most precise translation in the programme, because it turns a structural property of its central object into a classical analytic condition; the multiple-zero case, where leading coefficients suffice in A_p [BT, Theorem 4], is the natural next question.

9.2 The two-line system, Weil's explicit formula and Turing's method

Maps: $f \mapsto (u, v)$, carrying Q_A to $2\Omega(u) - 2\Omega(v)$; the loop $T^{\{-1\}}A$ to Weil's reflection $\#$; the holonomy count to $(N - V)/2$. *Established:* §4. *Suggests:* for Weil's side, a test family $\{f_k\}$ that is T-even by construction, with the explicit prime side $-G'/G$; for Turing's side, that an index for off-line zeros alone needs a map separating off-line pairs from multiple zeros. My assessment: the bridge recasts Weil's positivity as a signature statement about a form that is hyperbolic for every configuration of zeros; since the whole form has the same signature with or without RH, I expect progress along this bridge to come from the diagonal family, where the positivity question is concentrated (Question 2).

9.3 The programme's weights and Deligne's Weil II

Maps: $\rho \mapsto p^\rho$ of weight $2\operatorname{Re} \rho$; even tensor powers $\mapsto$ the positivity step of Weil II 1.5; the timed-prime measure $\mapsto$ Deligne's positive measure; $P_b \mapsto$ a relation $\operatorname{FNF}^{\{-1\}} = Q^{\{-1\}}N$. *Established:* §5, and Weil II §2.1 specialized to ζ is the classical proof of $\zeta(1 + it) \neq 0$ (DB9, DR0–DR4). Every solution of the relation on a jet block forces the weight shift $-2j$ of the programme's regrading, and such relations have solutions for every nilpotent, so what is established is a comparison of relations; the regrading, under which a jet block of multiplicity m satisfies the conclusion of 1.8.4 with centre $2\operatorname{Re} \rho - (m - 1)$ and is mixed in Weil II's sense [TY], is chosen in the programme rather than derived from an arithmetic source. Kurokawa's tensor products [Ma, §1.4] and Connes' square of the arithmetic site [CoE] are analogies for $H^1 \otimes H^1 \hookrightarrow H^2$. *Suggests:* for Weil II, a purity criterion and corrected lemmas; for ζ , the weight question (Question 8), which for the square is Manin's question of absolute Descartes powers of $\operatorname{Spec} \mathbb{Z}$ [Ma]. My assessment: the even-power computations supply the Euler-product half of Deligne's §1.5 mechanism, and the other half is the uniform pole bound; since that bound is equivalent to $\max \operatorname{Re} \rho \leq 1/2$ on finite exponent data, I think the decisive work here is an independent source for it, starting from the programme's own statement (TWC9, TWC11) that its computations leave room for one.

9.4 Nyman-Beurling in three topologies

Maps: the cover discrepancies in $\mathcal{B}$; two-sided dilation tests in $L^2(0, \infty)$; one-sided tests in $L^2(0, 1)$ and in Noor's Hardy space. *Established:* density in $\mathcal{B}$, unconditional (Proposition 3.11); two-sided density in $L^2(0, \infty)$ [Wi]; one-sided density equivalent to RH ([Be], [BD], Burnol's co-Poisson theory [Bu], [No]). *Suggests:* a comparison map between the settings, with its kernel identified, would show where the RH content enters. My assessment: the contrast is instructive, since the same family is dense unconditionally in one topology and exactly under RH in another; the sparse families of Question 5 probe which features of the topology matter.

9.5 Connes, Meyer and Connes-Consani

Maps: $Q \rightarrow$ Connes' Hilbert quotients [Co]; the doubled source $\rightarrow$ Connes-Consani's base Y ; the adelic periodization complex $C \approx \beta_{\text{pr}}[1] \oplus Q[0]$ (OMS7A.1); the absolute twistor line $[CCT] \rightarrow y^2 = x^3 - x$ through its $\mathbb{F}_{\{1^2\}}$ -points; the reversed prime knots of $[CCK] \rightarrow$ the conjugate character, with $W(\chi)W(\bar{\chi}) = 1$. *Established:* the kernel of the first map is exactly the off-line jets (§3.5); the second is a cohomology isomorphism (§7, S70) and the complex is formal (OMS7A.1); the twistor identification holds; the knot statements are theorems whose reading as anti-numbers is a proposal. Connes' §VIII distribution with the full Gram matrix is FGR, synthesis on Q is of Krasichkov-Ternovskii type [Kr], and the positive transfer forms are Bochner-Schwartz [VI]. *Suggests:* for Connes' approach, an exact account of what the Hausdorff completion loses and a jointly faithful family over all lines $\operatorname{Re} s = \sigma$ (VWR); for the programme, a dictionary with the established spectral realizations. My assessment: the comparison of Connes' Hilbert quotients with Meyer's Fréchet quotient is the cleanest structural statement here; the comparison of the classification of positive forms with Meyer's and the Weil-positivity literature has still to be made, and I would make it first.

9.6 Clocks, Bost-Connes, Beurling and monoids

Maps: the clock operators $S_n = \sqrt{n}V_n$, with $S_m S_n = S_{\{mn\}}$ and $S_m * S_n = S_{\{n/d\}} S_{\{m/d\}}^*$, $d = \gcd(m, n)$, the relations of $C(\hat{\mathbb{Z}}) \rtimes \mathbb{N}^\times [LR]$; timed primes $\mapsto$ Beurling's $D = \exp_*(R)$; the sheets $a =$

$1/q \mapsto$ congruence monoids. *Established*: the global quotient of all prime clocks is the Bost-Connes space (P14), whose partition function is $\zeta(\beta)$, with its phase transition at $\beta = 1$ [BoC] (its link with Proposition 6.1 is an interpretation, for which a map remains to be constructed); the timed-prime reconstruction is Beurling's construction on the actual primes (TP9-TP13); the Eulerian tests are the Krull-monoid criteria ([BC, Ca]; §2.3). *Suggests*: for monoids, a freeness test for every monoid of integers (Theorem 2.4); for ζ , that regularity of the counting function alone does not force RH (N-Beurling). My assessment: the monoid results are of independent interest as elementary number theory, Theorem 2.4 and Corollary 2.5 most clearly; their novelty was checked by one search only.

9.7 Two-prime density and mean-periodic functions

Map: the density of $(\log p)\mathbb{Z} + (\log q)\mathbb{Z}$ in $\mathbb{R}$, on which MCL3.4, IH6 and PL3 rest, $\mapsto$ the two-period phenomenon of mean-periodic functions [Sc]. *Established*: the density holds because $\log p/\log q$ is irrational; the link is a comparison of mechanisms. *Suggests*: it explains why two primes determine every zero while one prime leaves the aliases $\rho + 2\pi i k/\log p$. My assessment: spectral synthesis for mean-periodic functions is well developed, and transferring it to the programme's jet calculus is a direct and checkable next step.

9.8 The anomaly dictionary (a proposal)

The objects and the facts proved about them hold without assuming RH; the matching is a proposal.

Author's term	Exact object
anti-number	$g \mapsto \text{conj}(g(1/u))/u$ (Mellin shadow $s \mapsto 1 - \bar{s}$)
chirality	the mirror $A(s) = -\bar{s}$ through the pole at 0
the doublet	the orbit $\{\rho, \rho^\#, \rho^\# - 1, \rho - 1\}$ of an off-line zero
$\mathbb{Z}_2$ anomaly	the two lines as sheets of one double cover, deck transformation the central -1 of $SU(2)$
$\mathbb{Z}_1$ anomaly	the root-number phase, cancelled by the anti-sector: $W(\chi)W(\bar{\chi}) = 1$
inflow	the bulk between the lines, holding the zeros with $\text{Re } \rho < 1/2$ and their mirrors
the support τ	the pivot of the swing between 0 and 1: the reflection $s \mapsto 1 - \bar{s}$ about $1/2$

In these terms RH has three exact forms: every zero is its own anti-number; the bulk is empty; Weil's functional is nonnegative on every annihilation $g * g^\#$ (whose value at the unit, $\int |g|^2 du$, is positive with or without RH). My assessment: the dictionary is useful because every entry is an exact object; its test is whether a statement first seen in the physical language yields a new statement about these objects.

9.9 The $\mathbb{F}_1$ context

The author's description of τ as carrying $\mathbb{Z}_1$ and no $\mathbb{Z}_2$ is realized at Connes-Consani's generic point η , which every homeomorphism fixes and where no exchanged label can sit, although the sign ε survives there; the absence of $\mathbb{Z}_2$ thus has two exact readings, no exchanged label (true at η) or no sign ($\varepsilon = 1$, which among field-valued points holds exactly in characteristic 2). The labels $\mathbb{Z}_n$ follow the principle that an $\mathbb{F}_{\{1^n\}}$ -structure is a free $\mathbb{Z}/n$ -action ([CCM, §3]; a reading); $\mathbb{P}^1(\mathbb{F}_{\{1^2\}}) = \{0, \infty, \pm 1\}$, with the double cover $y^2 = x^3 - x$ branched there; the Eulerian even sheets are those with $(\mathbb{Z}/q)^\times = \{\pm 1\}$, the units of $\mathbb{F}_{\{1^2\}}$ (Proposition 2.8; the reading as $\mathbb{F}_1 \rightarrow \mathbb{F}_{\{1^2\}}$ is an interpretation); and $G(\mathbb{Z})$ adds a Boolean point below the generic point of $\text{Spec } \mathbb{Z}$, where Connes-Consani add a closed point ∞ (a reading). My assessment: the reading with mathematical content of its own is the coincidence of the Eulerian even sheets with the units of $\mathbb{F}_{\{1^2\}}$, which deserves a statement independent of the reading.

9.10 The author's other workbenches

Companion records treat the Erdős-Straus (ES), Yang-Mills (YM, with the Navier-Stokes (NS), S^6 and Jacobian material), Erdős Problem 817 and Collatz workbenches. Edges located from the zeta side:

Edge	What crosses	Status
NS → ES	the torus endomorphism $[[3, 1], [1, 5]]$; exact aliasing on finite character groups	verified; joined
NS → YM	NS velocity as a reducible $SU(2)$ connection, $-2\sum \text{tr } F_{ij}^2 = \lambda^2 \text{curl } u ^2$	verified, confirmed
zeta → YM	the Gram sandwich (§6)	verified
Jacobian → NS → YM	$U = DF^{-1}(V \circ F)$, the escaping curve, YM Theorem 14.2	verified (Lemma 14.2)
$S^6 \rightarrow YM$	the imaginary period block, through $D = \det B$	verified
$ES \leftarrow \text{zeta}$	the semiring $G(R) = R \sqcup \{\tau\}$; a Keller map P with $\det DP = -2$	verified; for τ
$S^6 \rightarrow ES$	the $(3, 4, \infty)$ monodromy covers	verified at ∞
Collatz $\leftrightarrow ES \leftrightarrow \text{zeta}$	the homogenization $(x, h) \mapsto (ax + bh, h)$	verified in part
817 → zeta	finite word sources in the supported quotient; a cocycle identity	one of ten items
zeta → ES, NS → zeta	theorem dependencies (NG20-NG21); Mellin transports	not verified

One Collatz item is verified here: with $T(n) = 4n + 1$, $\theta(n) = 3n + 1$ and the odd step $U(n) = (3n + 1)/2^{\nu_2(3n+1)}$, $\theta(T(n)) = 4\theta(n)$, so $U(T^j n) = U(n)$ for all j . My assessment: the shared linear algebra (Gram sandwiches, Schur complements) is the most reusable crossing, since it is exact and needs no identification of objects; the edges that carry a statement about ζ into another workbench, or back, still need typed maps with verified content.

10. Questions

- The primary decomposition.** Are the principal parts of $1/\zeta$ at the zeros polynomially bounded (Theorem 3.9(c))? For multiple zeros, do the leading coefficients suffice?
- The diagonal family.** Is the T-even family $\{f_k\}$ of Corollary 4.7 easier to control on the prime side, through $-G/G$, than Weil's own family?
- An index for off-line zeros.** Is there an index, from a map on the strip, that counts off-line zeros alone (Proposition 4.9), and can one-line inputs bound the bulk count better than Turing's method?
- The Davenport-Heilbronn function.** Does the two-line splitting fail for it?
- Sparse degrees.** Is the family with $n \in \{2^a 3^b\}$ dense in $\mathcal{B}$ for $0 < t \leq 0.06060$? Carleman's formula in the sectors $|\arg w| < \pi/(2k)$, applied to $L(z^{\{1/k\}})$, gives the thresholds $(\log 2)(\log 3)/(2\pi k)$ for $0 < k \leq 2$ (from $(4 - k^2) \int_{-\alpha}^{\alpha} \sin^2 \varphi \cos k\varphi d\varphi + 2k \sin^2 \alpha = 4/k$, $\alpha = \pi/(2k)$; check Z7), while for $k > 2$ its zero sum converges and step 3 gives no contradiction; a Jensen argument needs $t > (\log 2)(\log 3)/2 \approx 0.381$. Which sets S of quadratic counting give density for every $t > 0$?
- Velocities and Beurling systems.** Do the one-sided statements for the sharp velocity sums hold, and does the sharp spectrum differ from the Dirichlet spectrum off the Eulerian sheets (suggested numerically at $a = 2$)? Does $N(x) = cx + O(x^\theta)$, $\theta < 1/2$, force $\zeta_P \neq 0$ on $\sigma > 1/2$?
- The local model at τ .** Which local model of the test functions at τ does the programme intend (Proposition 6.1)?
- The weight question.** Can the reconstructed arithmetic supply a square with a Künneth map and discrete weights, or an independent source for a pole bound uniform in the tensor degree (§5.1)? A ramified map reported in the programme's sessions of 24–25 September, with an exact factor n^{-1} in the nilpotent relation, is to be compared with this question.

11. Verification

Every numerical or symbolic claim of the source has a check script with recorded output; the scripts are in the record's checks/zeta/ folder. They use mpmath, sympy and, for argument-principle counts, Arb ball arithmetic; checks marked expected-fail document a finding (for example the counterexample of Lemma 5.4).

- The version-1 checks: 48 scripts at the top of checks/zeta/ (47 with a recorded _OUTPUT.txt), including F1–F9 and G1–G3 (Theorem 3.9), B1–B5 (Proposition 4.6), D1–D4 and R5 (Proposition 4.9), C1–C3 (Proposition 4.10) and B1–B4 (Lemma 5.4).
- The first referee pass: four scripts in checks/zeta/referee08/, with their argument-principle outputs.

- Five figure scripts in checks/zeta/figures/, and the code of §2.1 in checks/zeta/copy_round2_code/, which reads its zero data from the directory named by ZERO_DATA_DIR.
- Version 2: zeta_reader_addendum_checks.py (Z1–Z8), and in checks/zeta/zeta_reader_review/ the review’s ten scripts, the eleven scripts of verify_minors/ and the sixteen of referee_v2/.

Not read or not re-derived (not verified here): the products named in the final reports of the programme’s sessions of 24–25 September (among them GPC0–GPC12, CW1–CW8, TPL0–TPL13, BQC0–BQC6, ASQ0–ASQ12, CQF0–CQF10 and CCR0–CCR10); TD, GHE beyond GHE1, TP and S1–S7 (read in content maps only); NPE5–NPE10 and NER; GAP, GER, GEX, CTF, ASD, JTB, PGD, PGC, SCL, DPL, RSR and SSI10 (read in part); and, taken on trust, Meyer’s paper, CFPA.1–CFPA.7, GSP and CTS, and the Yang–Mills lattice calculations and fixed-box limits. Still to be done: a line-by-line re-derivation of Theorem 2.1, with checks at $a = 1/2$ and $a = 2$.

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